FLOE

Random-Access AEAD for Fast Lightweight Online Encryption

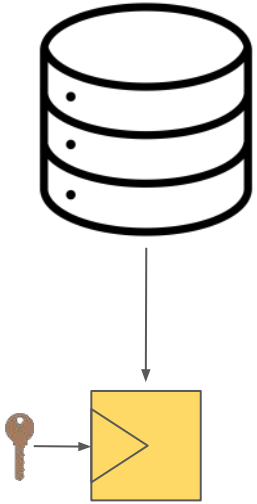
Andrés Fábrega **Greg Rubin** Julia Len Thomas Ristenpart

Cornell University snowflake The University of North Carolina at Chapel Hill UNIVERSITY OF TORONTO

To appear at Eurocrypt 2026 https://eprint.iacr.org/2025/2275

We encountered a problem…

- Encryption of large files
- Must use Authenticated Encryption (AES-GCM)



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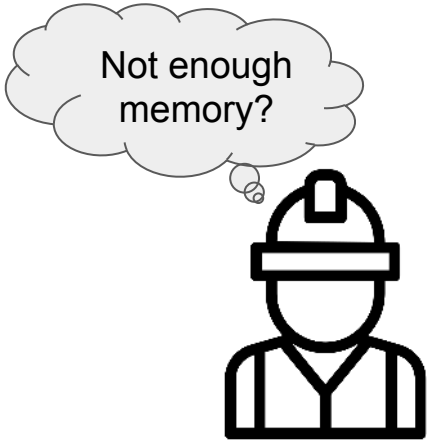

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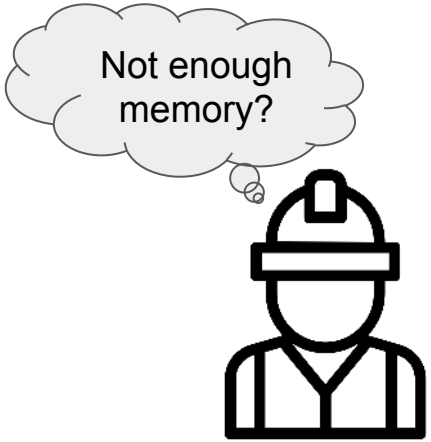

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Buffering that much data in memory causes problems.

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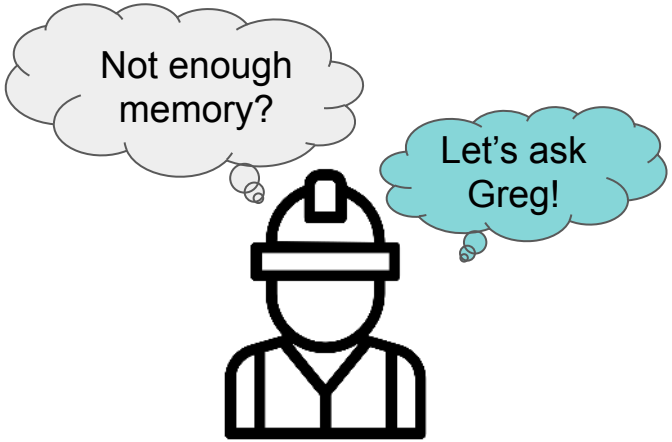

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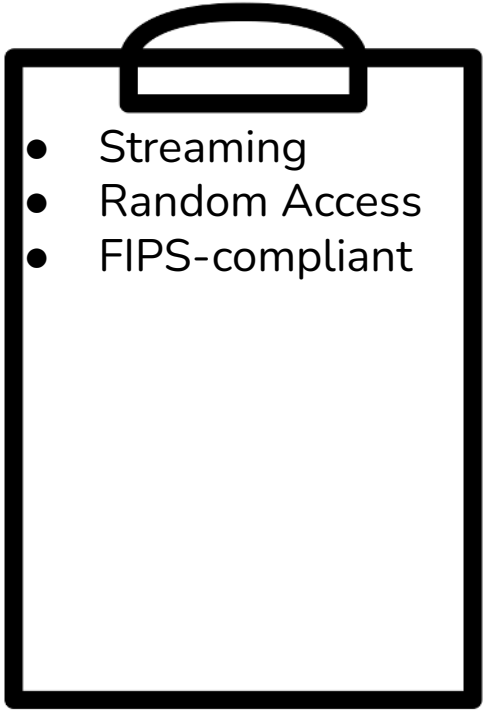

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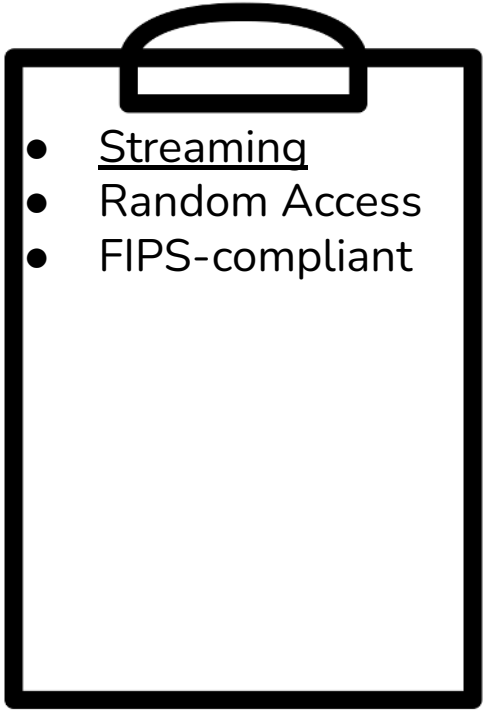
Streaming:
Can encrypt/decrypt large files sequentially with limited memory.



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- Streaming
- Random Access
- FIPS-compliant

Random Access:
Can encrypt/decrypt semi-arbitrary subsections.

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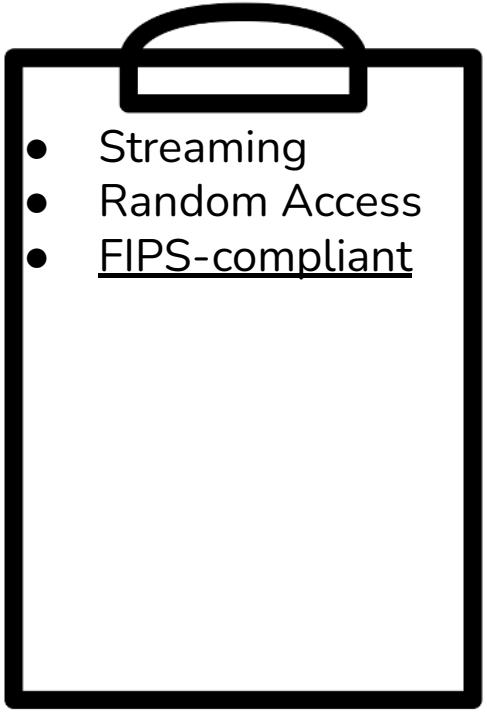
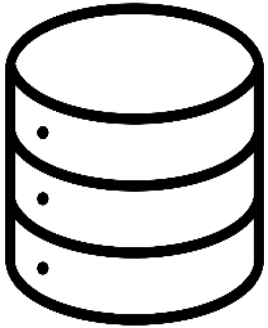
Streaming:
Can encrypt/decrypt large files sequentially with limited memory.

FIPS-Compliant:
Only uses approved algorithms (AES-GCM and HKDF_Expand/SP 800-108) in an approved manner (with random IVs).

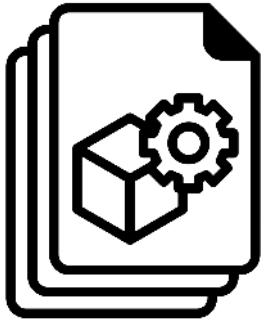


Random Access:
Can encrypt/decrypt semi-arbitrary subsections.

This is a common problem

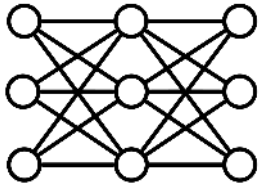


Databases



Machine Images

Container Images



Model Weights

And so much more...

Why not just break it up?

A	B	C	…	X	Y	Z

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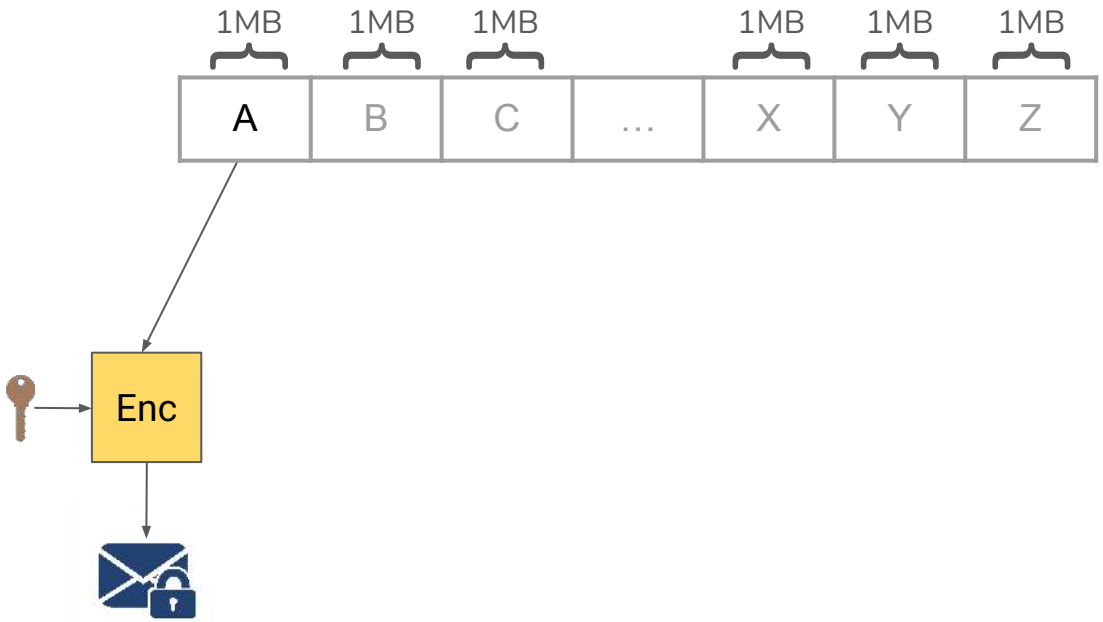

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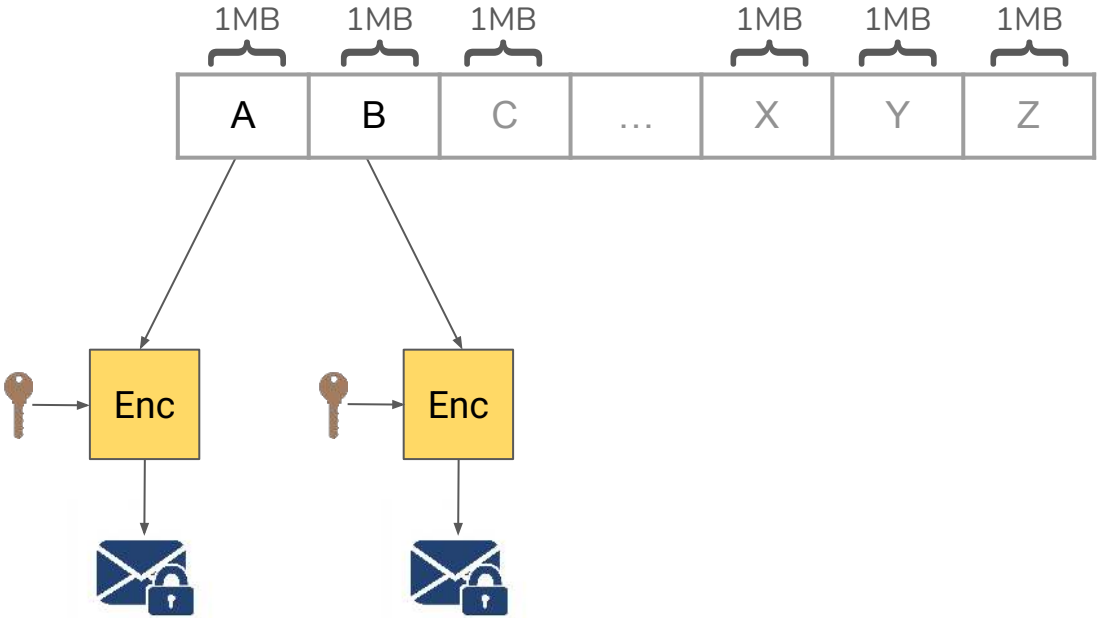

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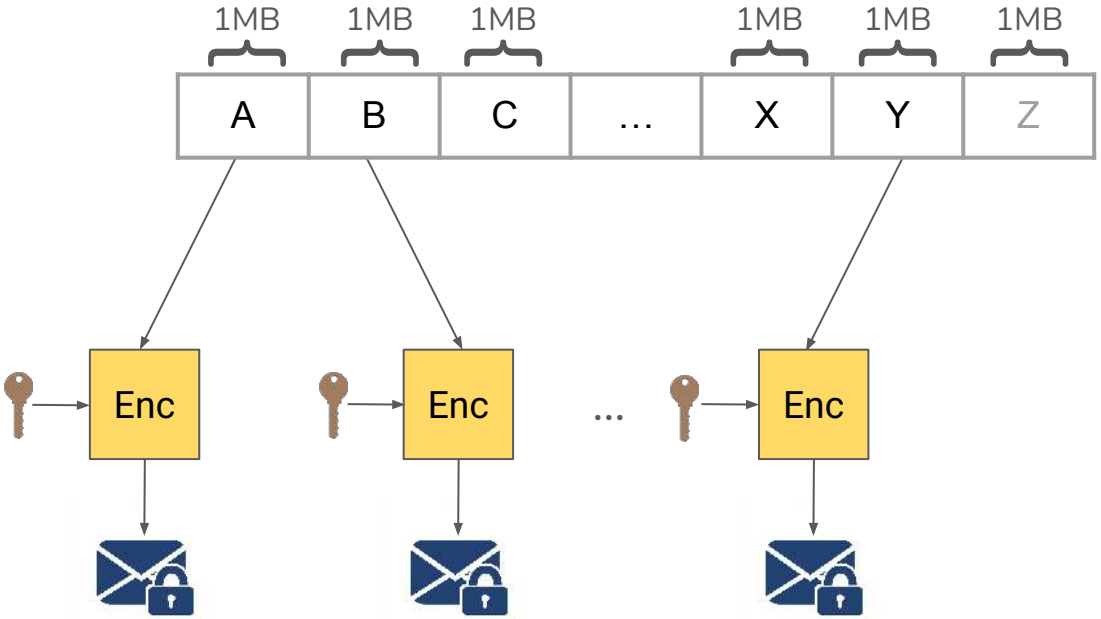

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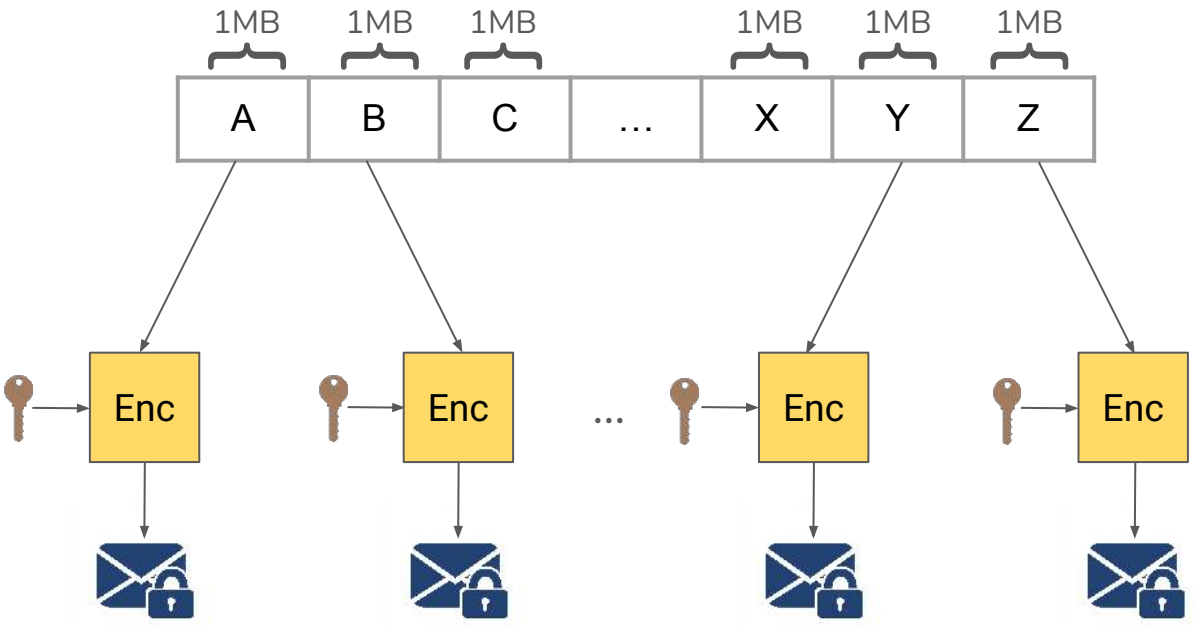

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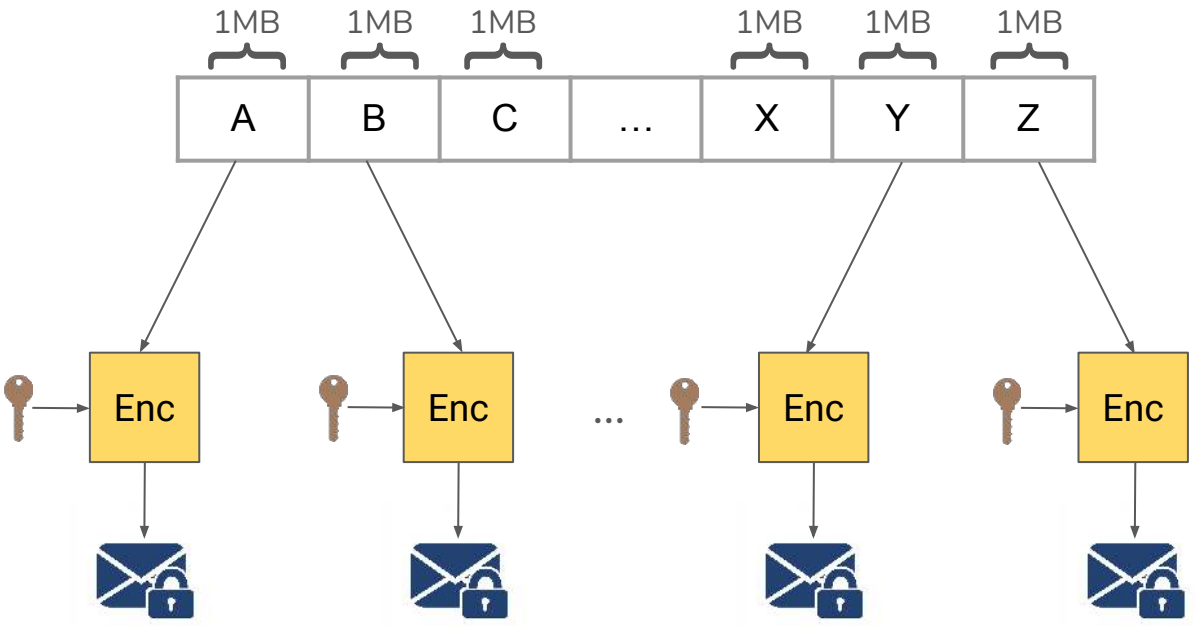

The obvious solution *almost* works…

Doesn’t protect against inter-segment attacks!

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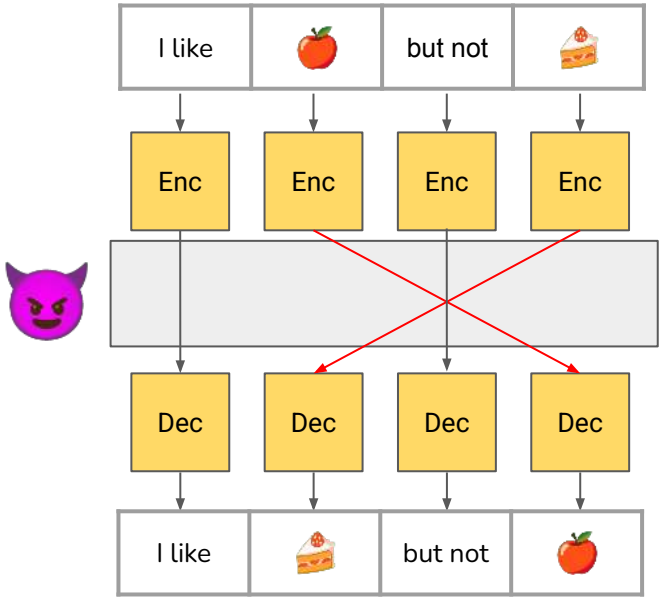

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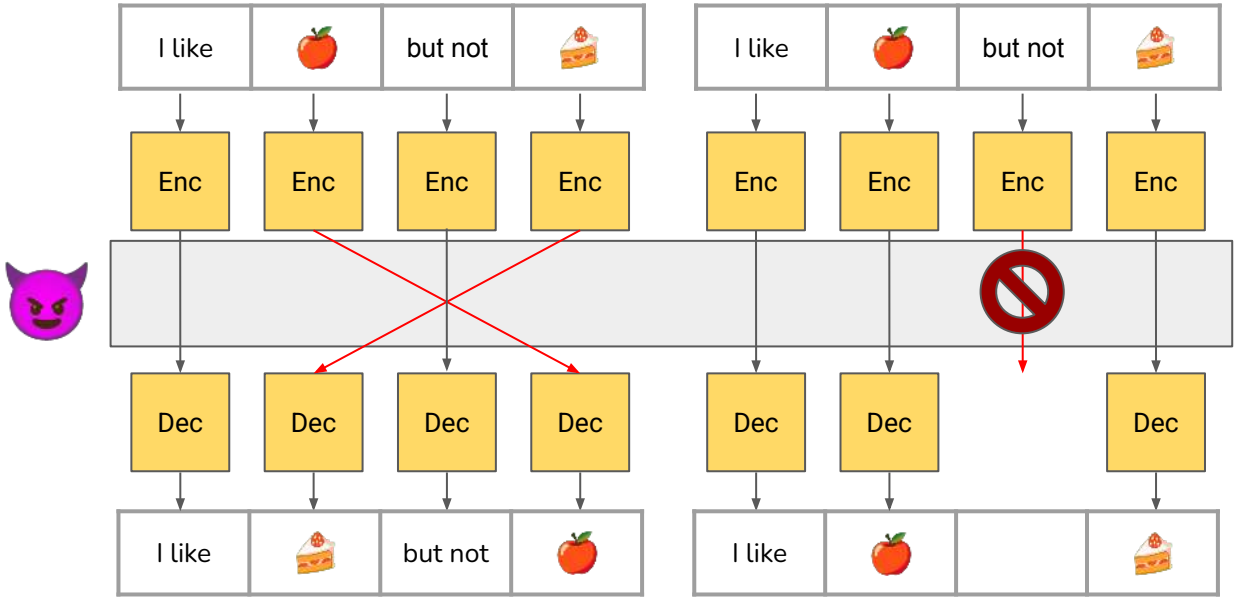

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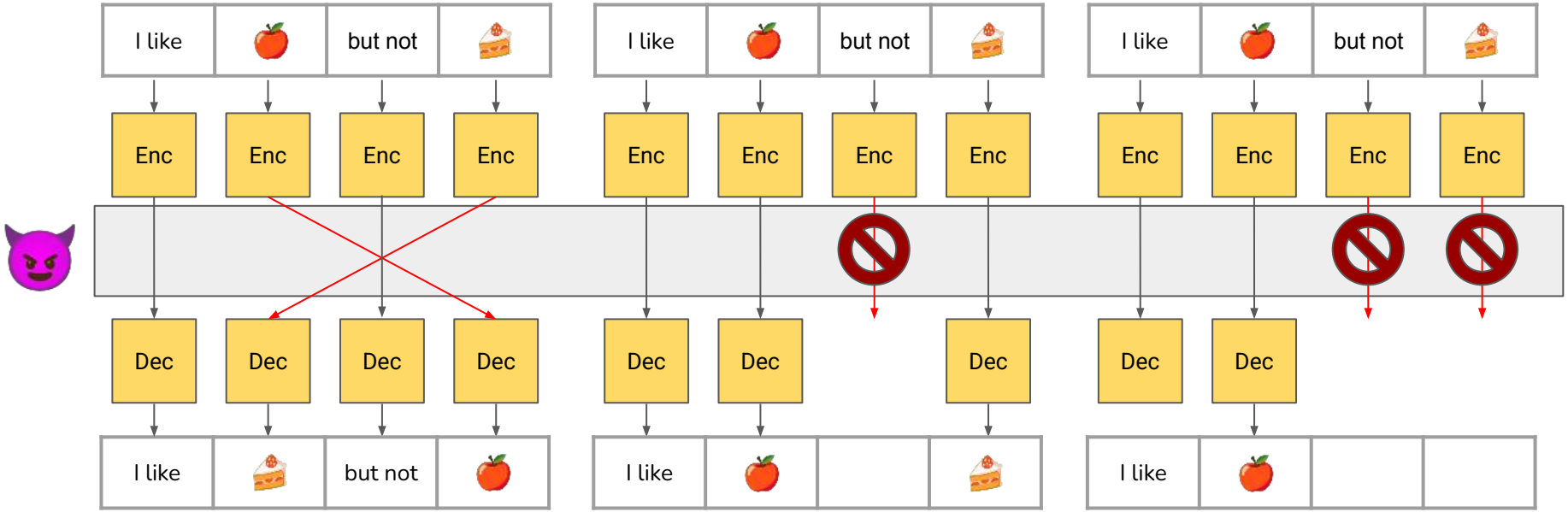


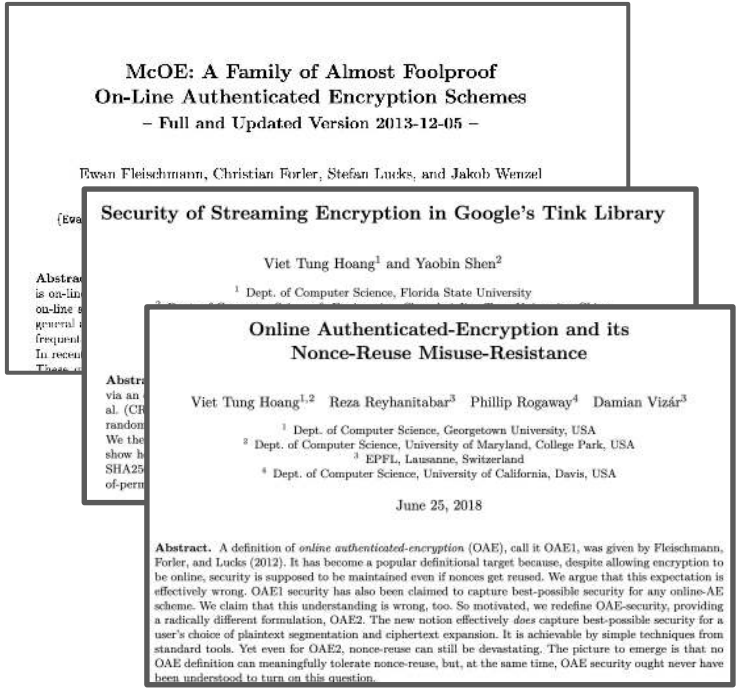
Solved Problem Right?

Literature review time!

Fleischmann et al. in 2012
Hoang et al. in 2020, 2015

Many encryption libraries over the years.

Google Tink



McOE: A Family of Almost Foolproof On-Line Authenticated Encryption Schemes
– Full and Updated Version 2013-12-05 –

Ewan Fleischmann, Christian Forler, Stefan Lucks, and Jakob Wenzel

Security of Streaming Encryption in Google's Tink Library

Viet Tung Hoang[1] and Yaobin Shen[2]

[1] Dept. of Computer Science, Florida State University

Online Authenticated-Encryption and its Nonce-Reuse Misuse-Resistance

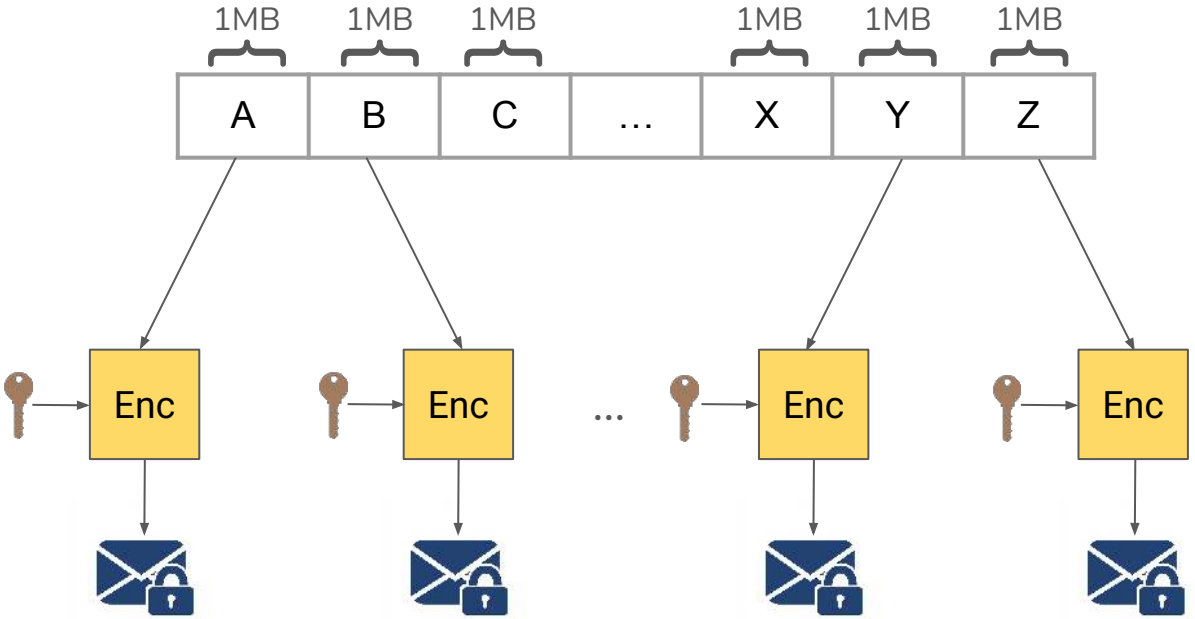
Viet Tung Hoang[1,2] Reza Reyhanitabar[3] Phillip Rogaway[4] Damian Vizár[3]

[1] Dept. of Computer Science, Georgetown University, USA
[2] Dept. of Computer Science, University of Maryland, College Park, USA
[3] EPFL, Lausanne, Switzerland
[4] Dept. of Computer Science, University of California, Davis, USA

June 25, 2018

Abstract. A definition of *online authenticated-encryption* (OAE), call it OAE1, was given by Fleischmann, Forler, and Lucks (2012). It has become a popular definitional target because, despite allowing encryption to be online, security is supposed to be maintained even if nonces get reused. We argue that this expectation is effectively wrong. OAE1 security has also been claimed to capture best-possible security for any online-AE scheme. We claim that this understanding is wrong, too. So motivated, we redefine OAE-security, providing a radically different formulation, OAE2. The new notion effectively *does* capture best-possible security for a user's choice of plaintext segmentation and ciphertext expansion. It is achievable by simple techniques from standard tools. Yet even for OAE2, nonce-reuse can still be devastating. The picture to emerge is that no OAE definition can meaningfully tolerate nonce-reuse, but, at the same time, OAE security ought never have been understood to turn on this question.

Previous Approaches: STREAM & Tink Streaming



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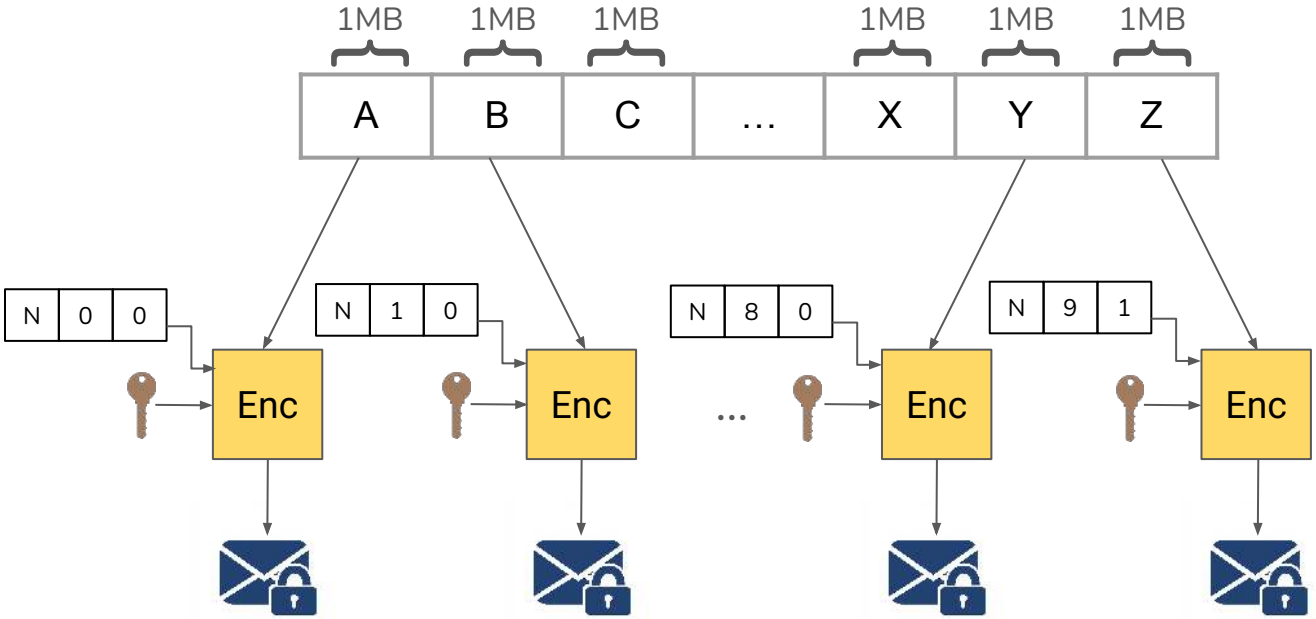

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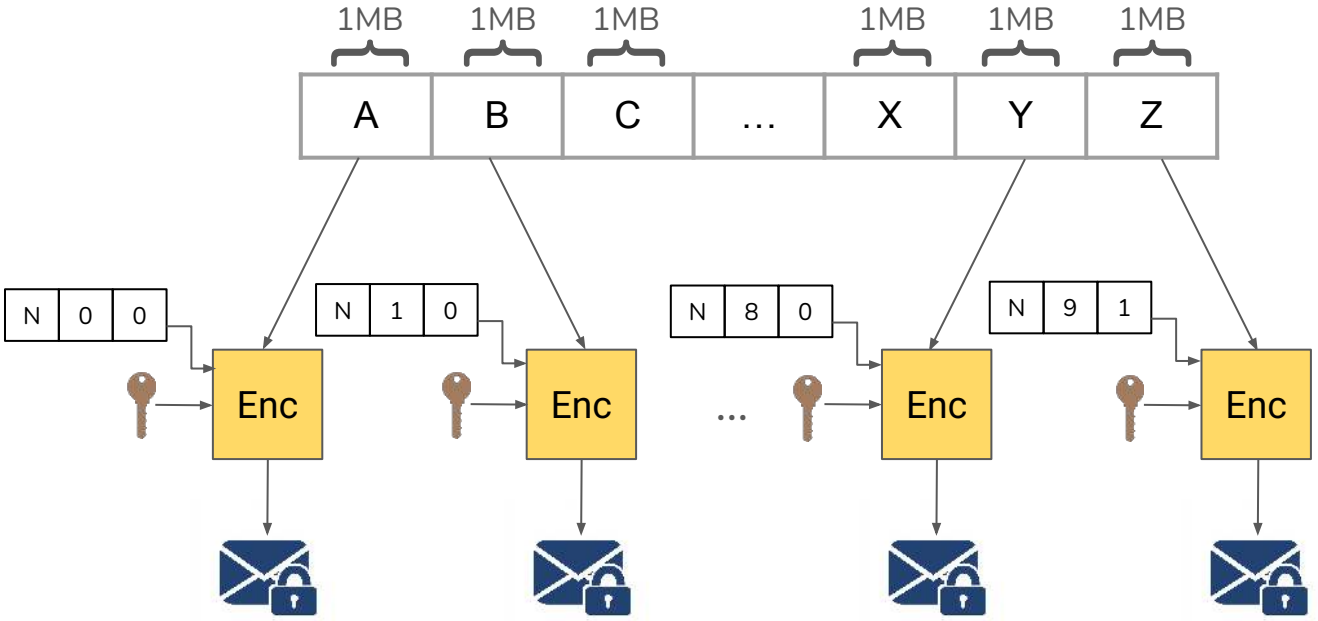

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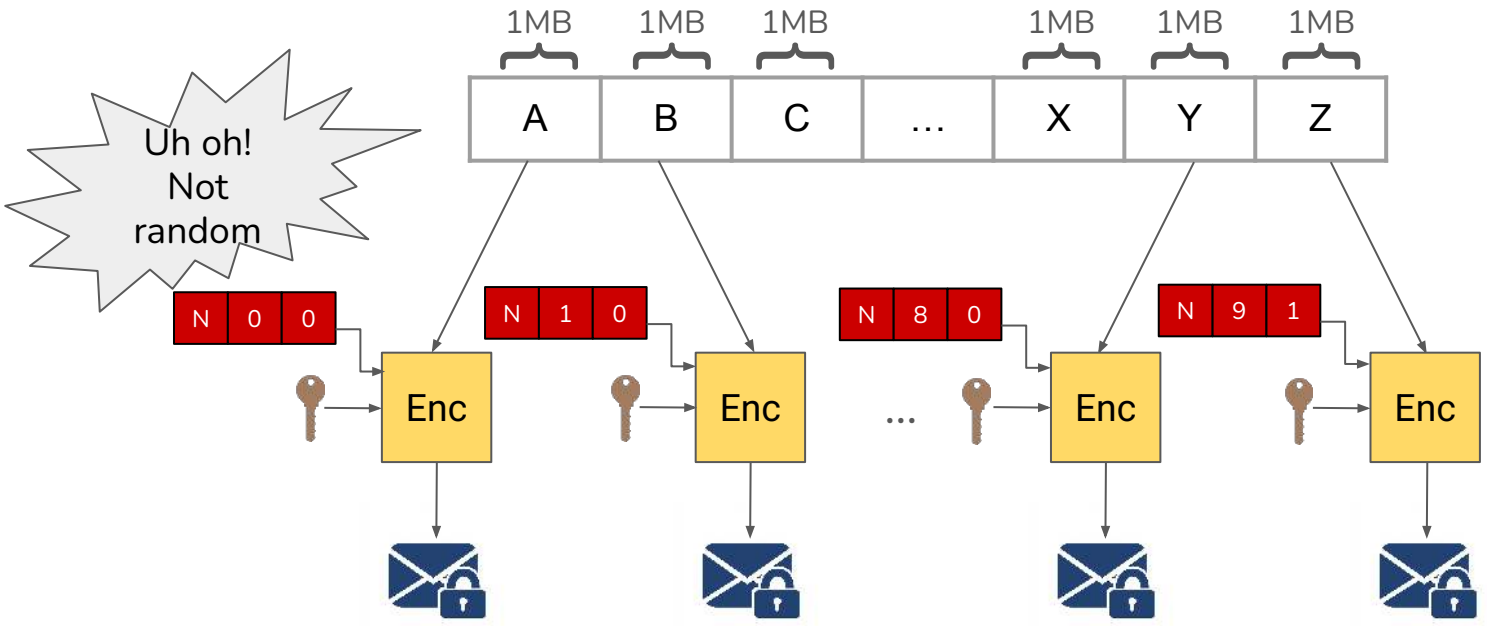

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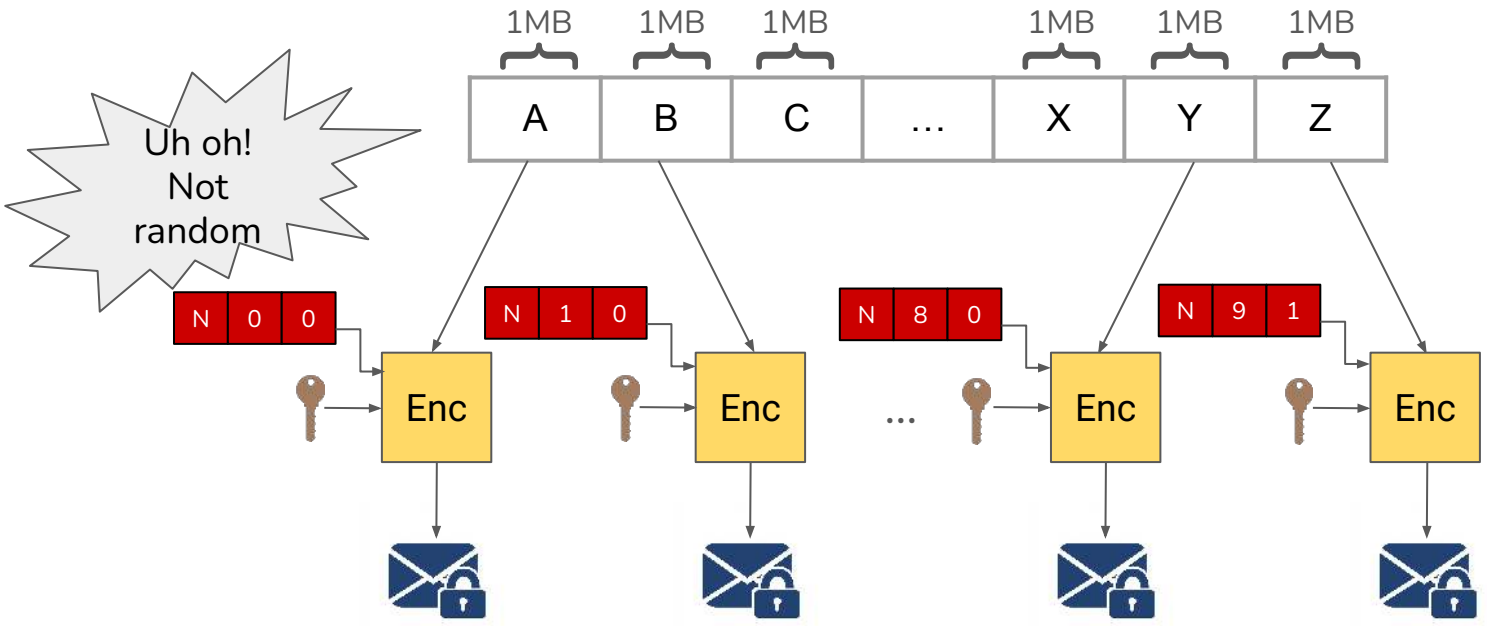

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Random-Access Encrypt is an open question…

We need something new

While we are at it… let’s add some desiderata!

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- FIPS-compliant

- Safe to use
- Easy to implement
- Safe error messages
- Committing

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Easy to use. Hard to misuse. Protects against common mistakes.

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Committing:
Modern additional security property. Ensures a single ciphertext can only decrypt to a single plaintext.

FLOE

Fast Lightweight Online Encryption

Photo by David Rubin
Used by permission

FLOE

Fast Lightweight Online Encryption

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- A low, flat mass of floating ice.

~ Wiktionary

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- A modern efficient encryption algorithm.

~ This paper

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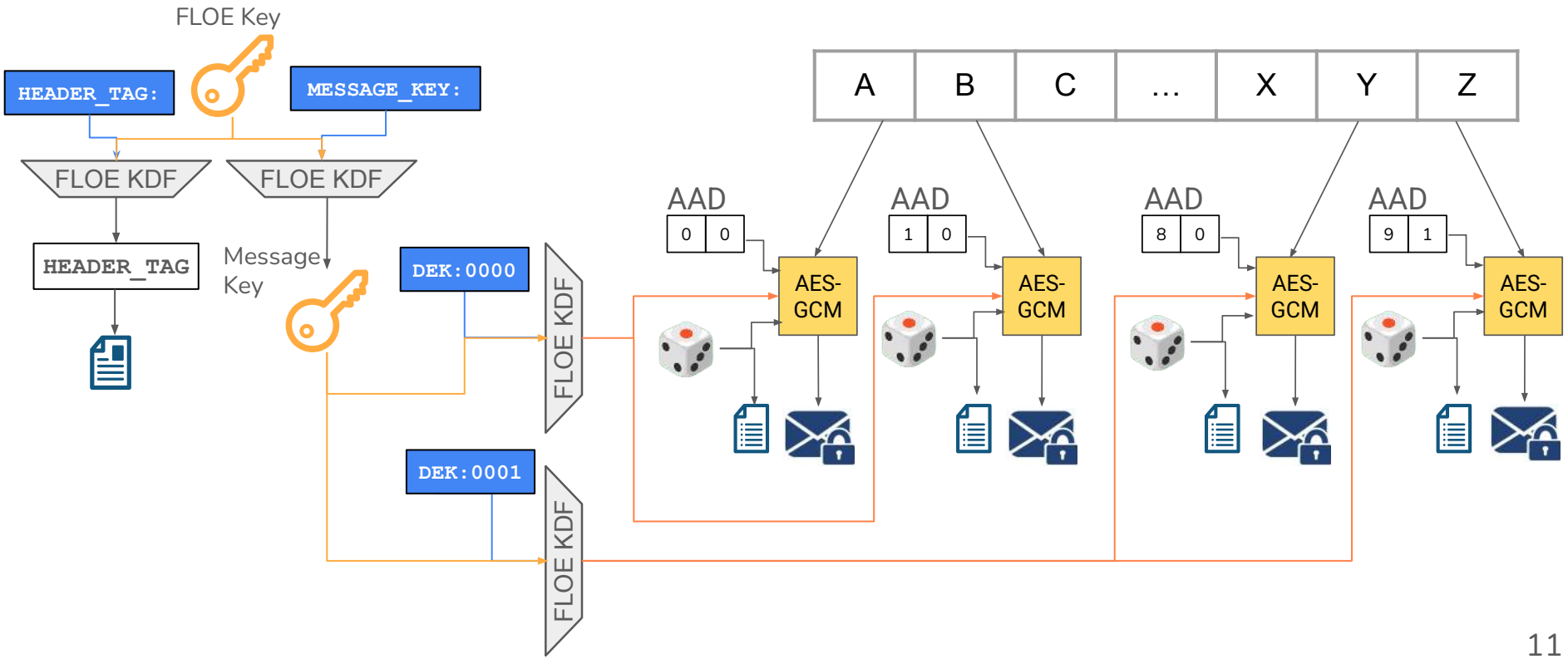
FLOE Key
HEADER_TAG:
MESSAGE_KEY:
FLOE KDF
FLOE KDF
HEADER_TAG
Message Key
DEK:0000
DEK:0001
FLOE KDF
FLOE KDF
A
B
C
...
X
Y
Z
AAD
0 0
AAD
1 0
AAD
8 0
AAD
9 1
AES-GCM
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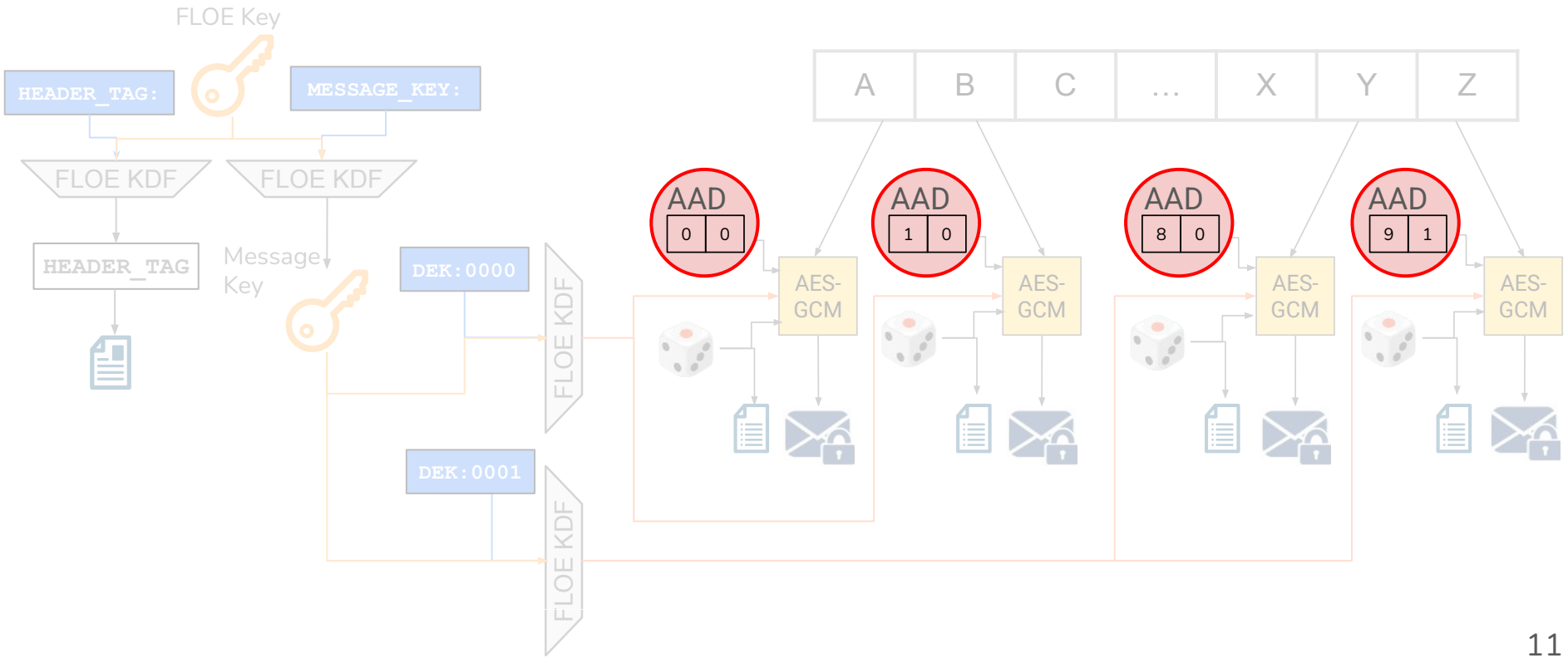
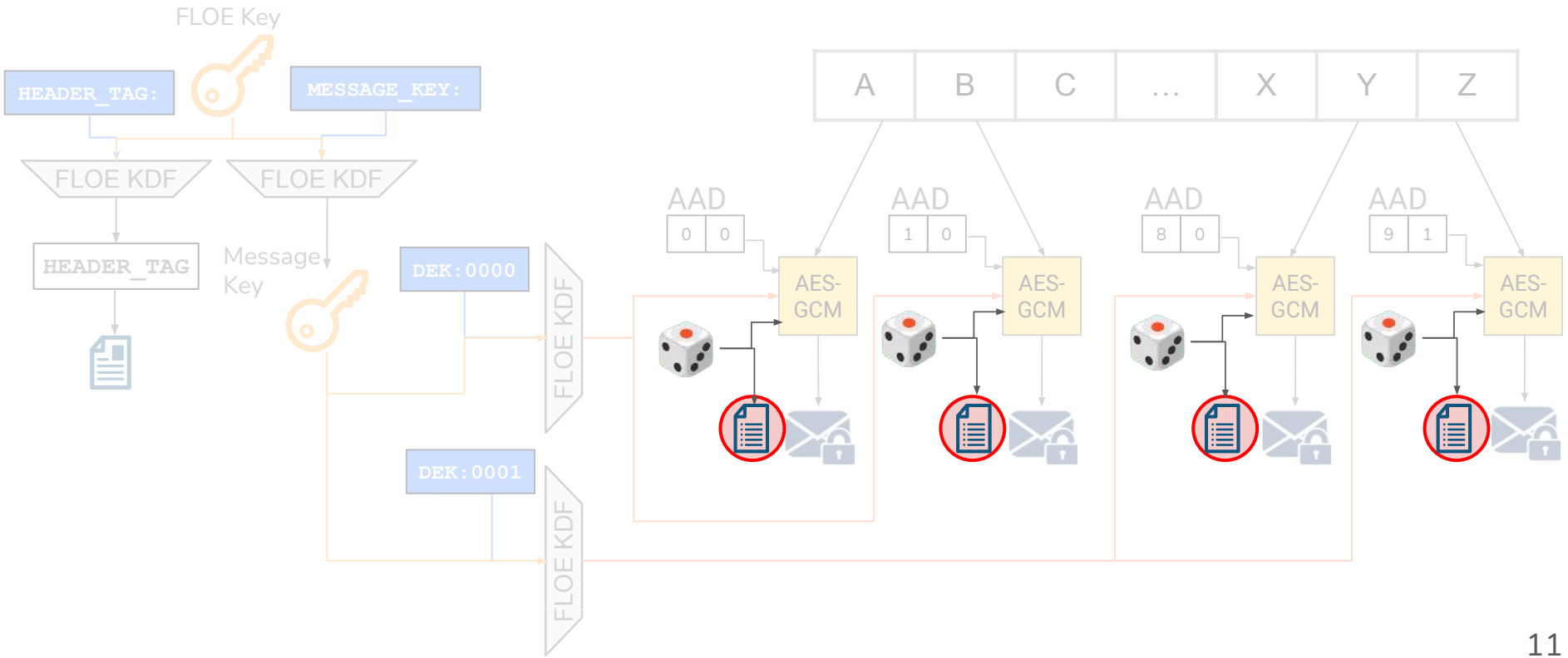
FLOE

Segment location is moved to the AAD



FLOE

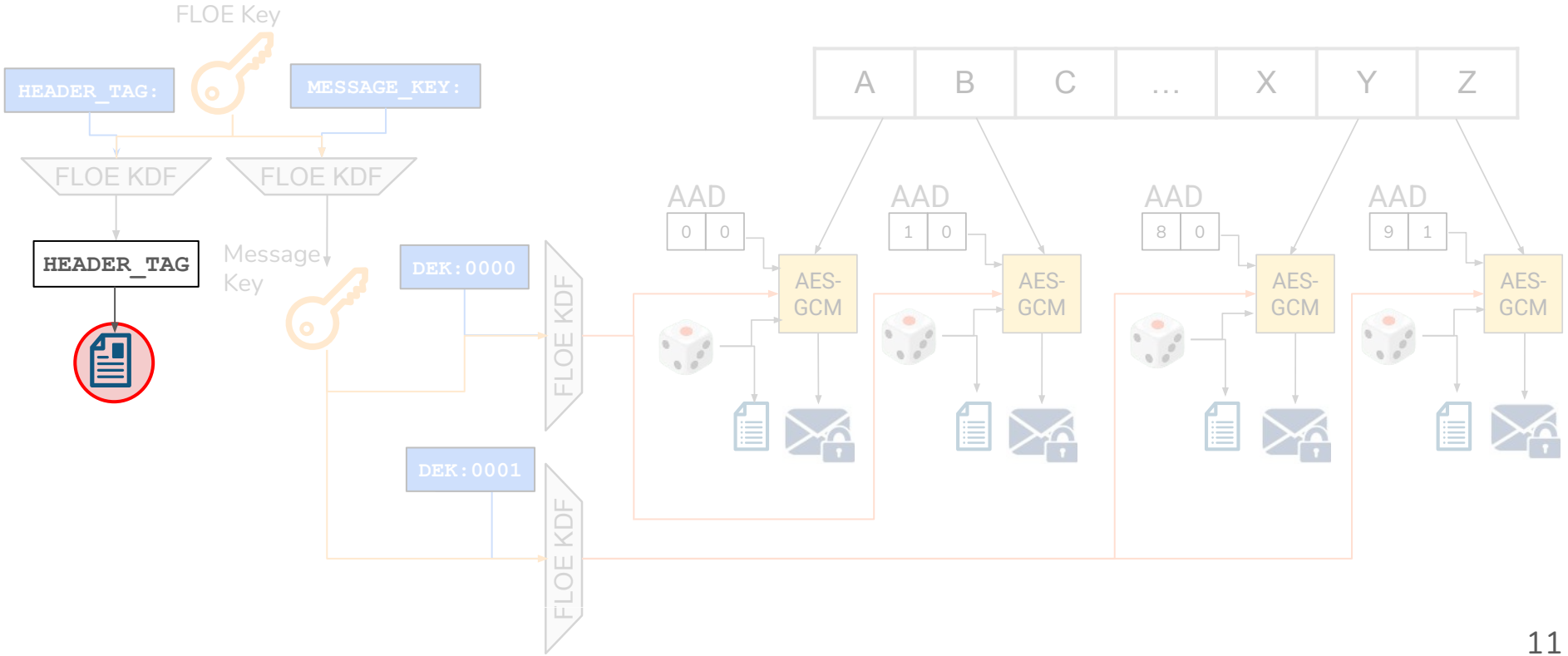
Segment headers to encode random IVs and simplify streaming decryption



FLOE

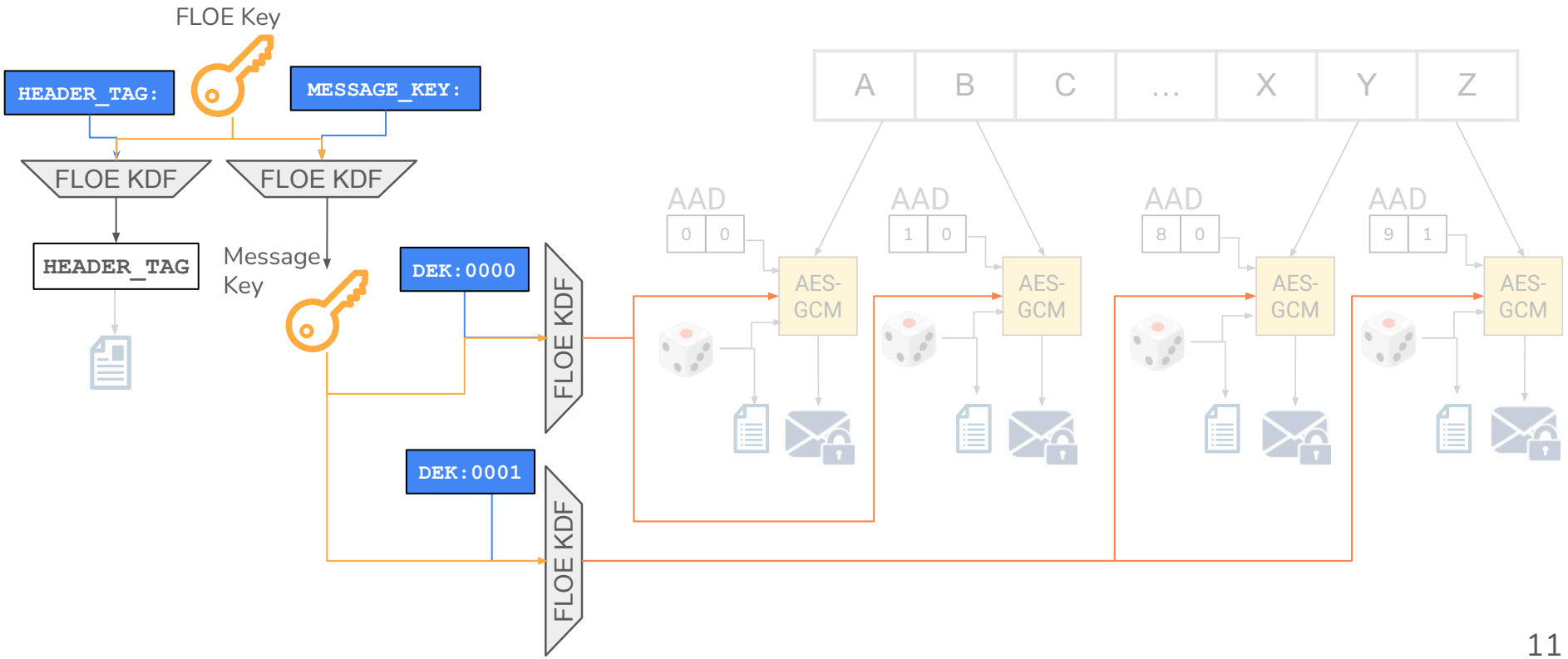
Ciphertext header for commitment, parameters, and main IV

AEAD	KDF	SEG_LEN	IV_LEN	IV	HEADER_TAG



FLOE

Key Schedule ties things together and supports exabytes under a single key



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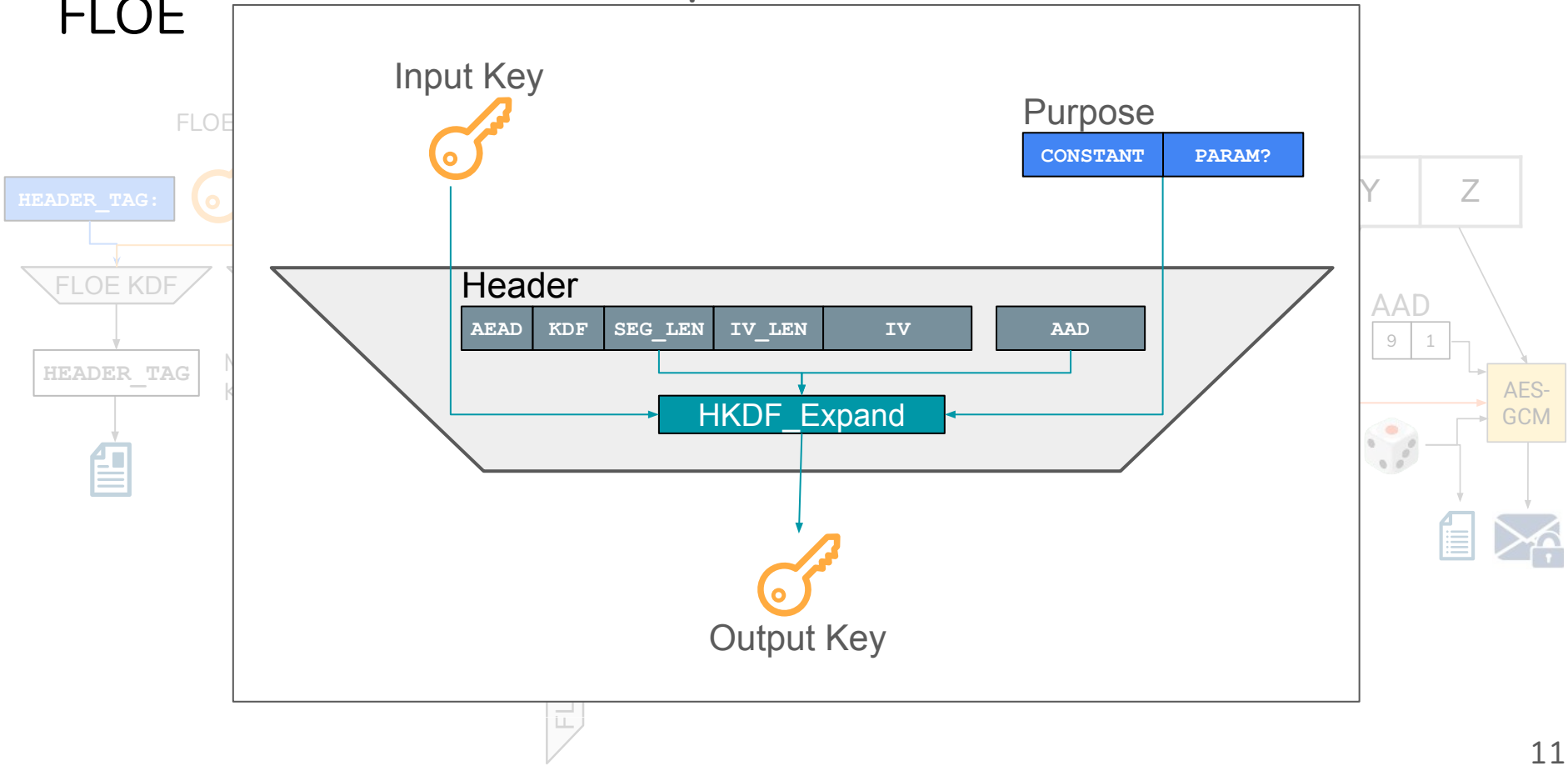
The KDF includes all parameters and main IV



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Is FLOE secure?

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1. In-order encryption and decryption (Hoang et al., 2015)
   a. Primitive: online AE scheme
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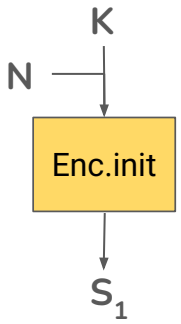

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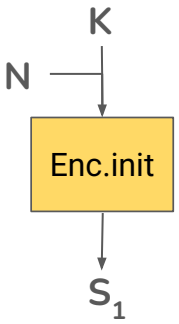

I like	🍎	but not	🍰

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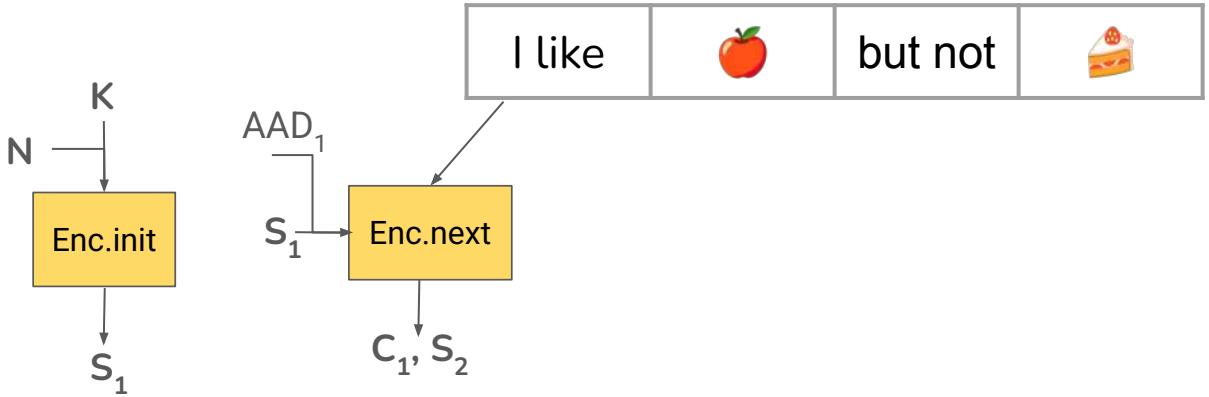

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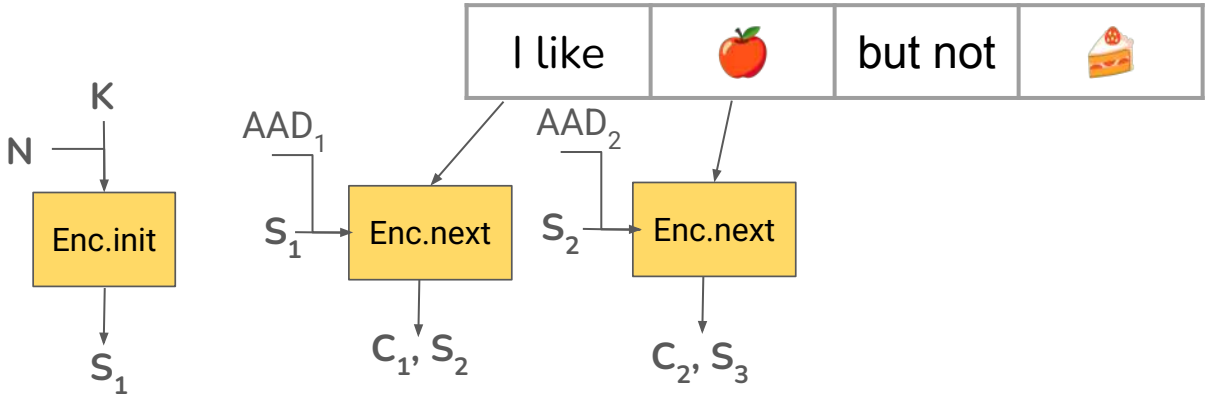

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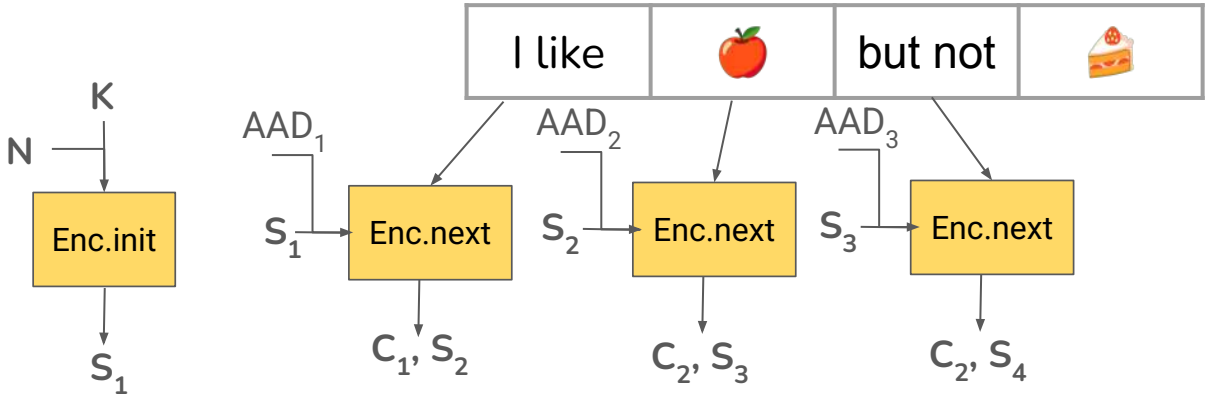

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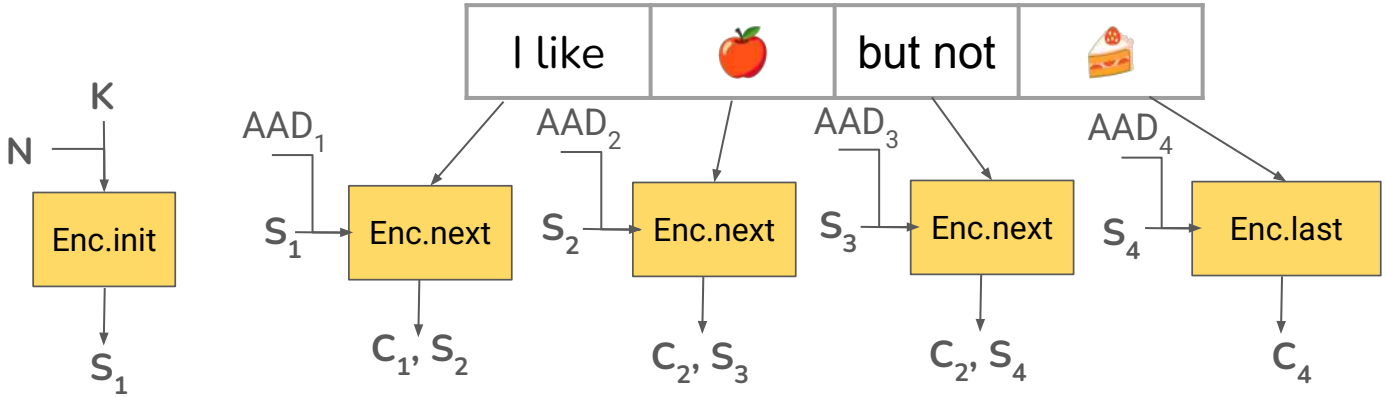

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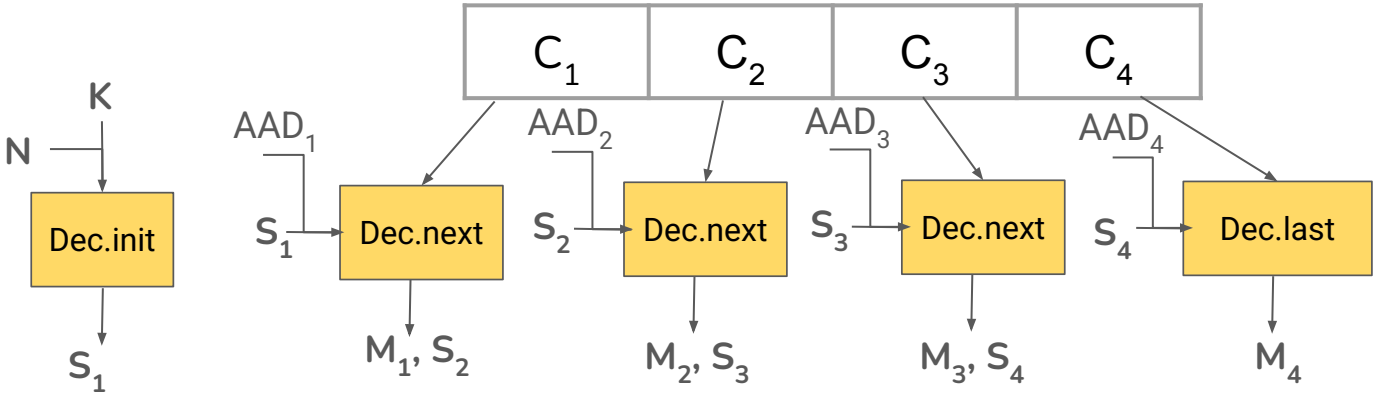

What does security mean in our setting?

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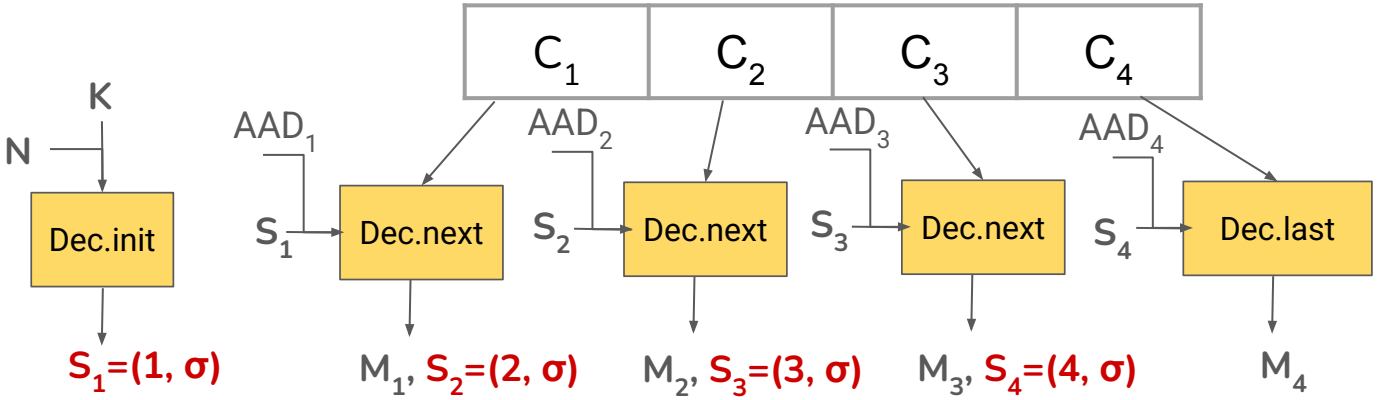

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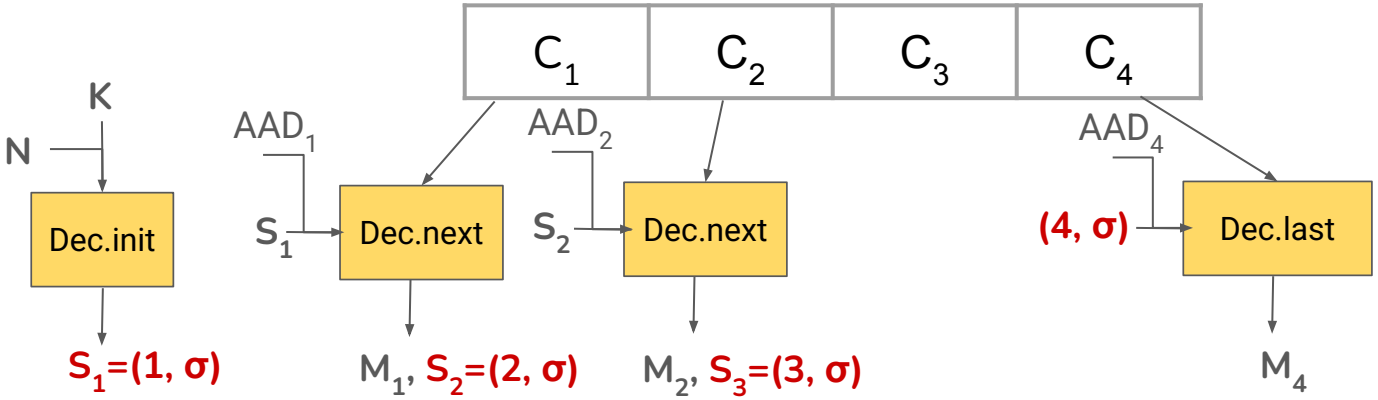

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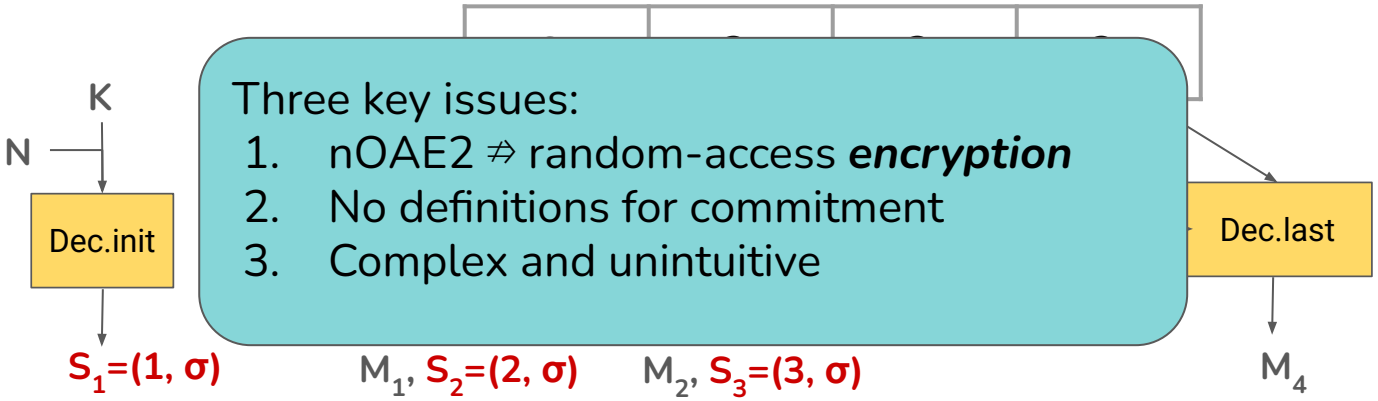

Three key issues:
1. nOAE2 ⇏ random-access ***encryption***
2. No definitions for commitment
3. Complex and unintuitive

New primitive: random-access authenticated encryption (raAE)

Single segment encryption/decryption process; encode positions explicitly

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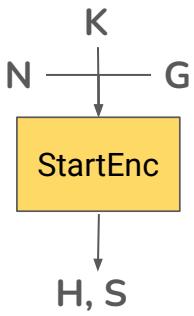

I like	🍎	but not	🍰

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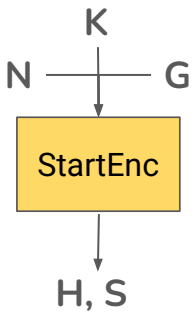

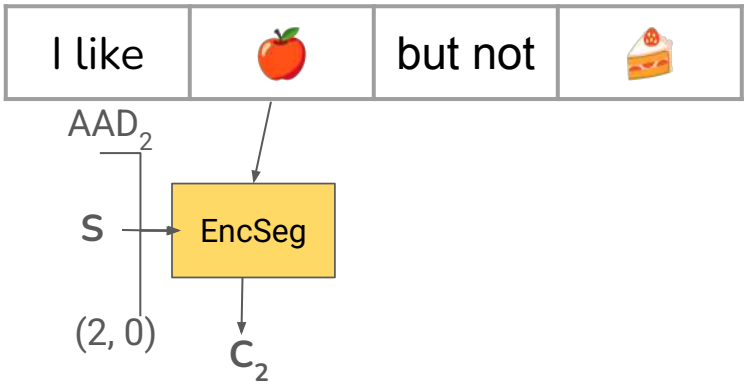

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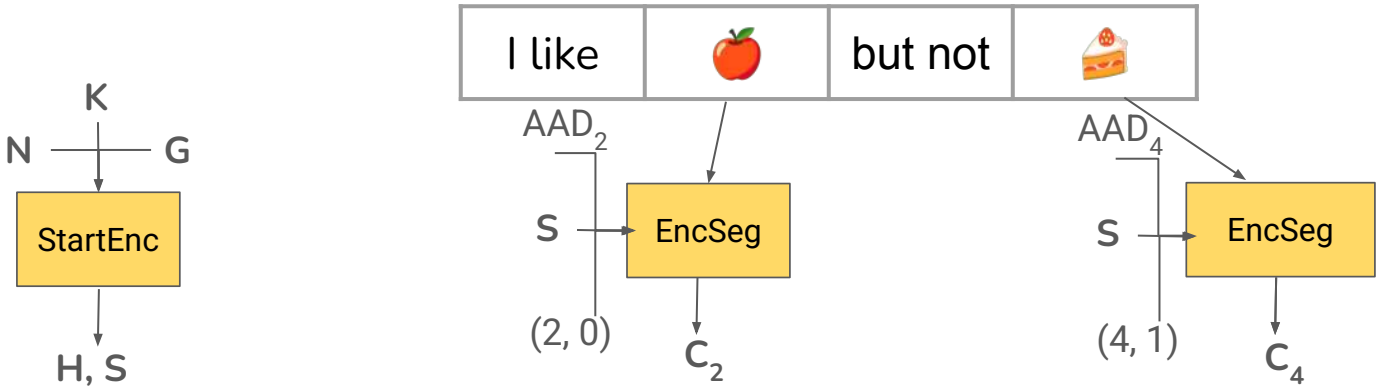

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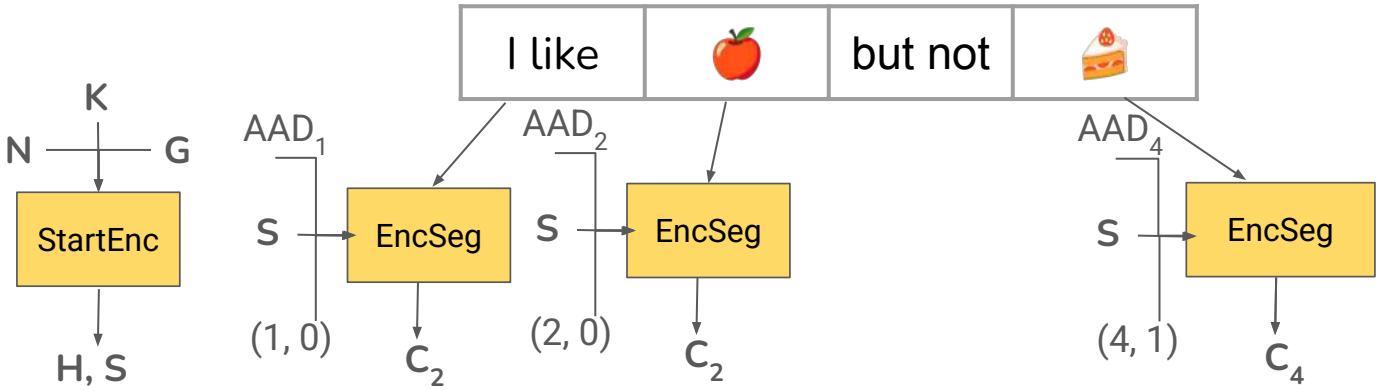

New security notion: random-access real-or-random (ra-ROR)

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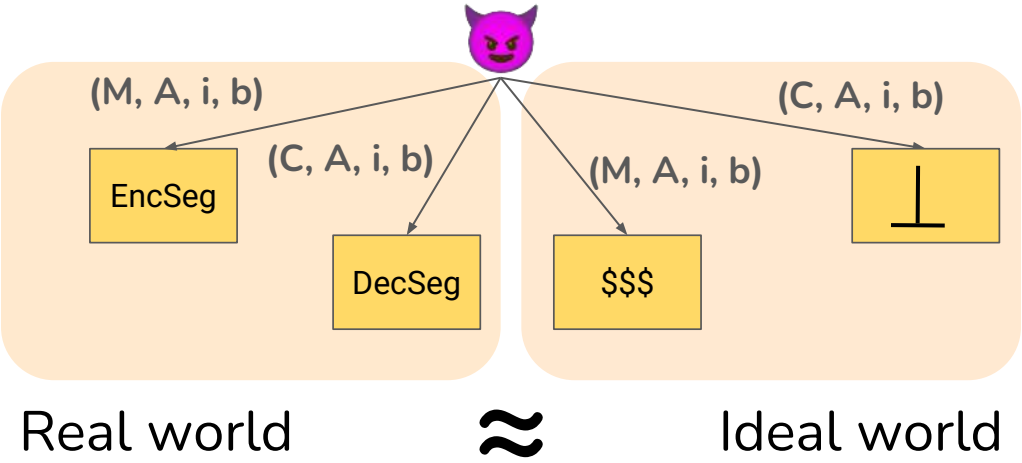

ra-ROR is strictly stronger than nOAE2

ra-ROR $\Rightarrow$ nOAE2
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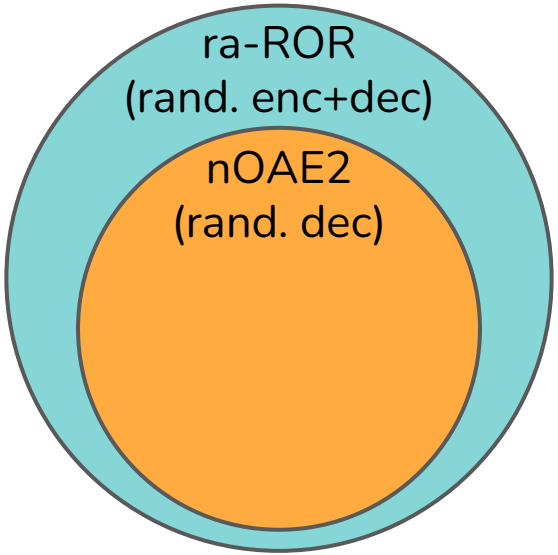

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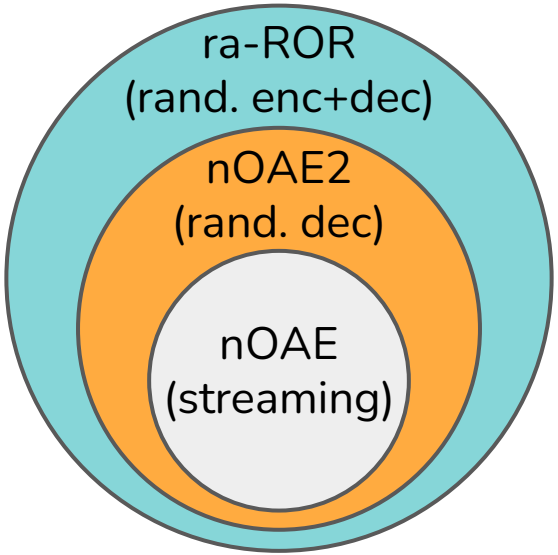

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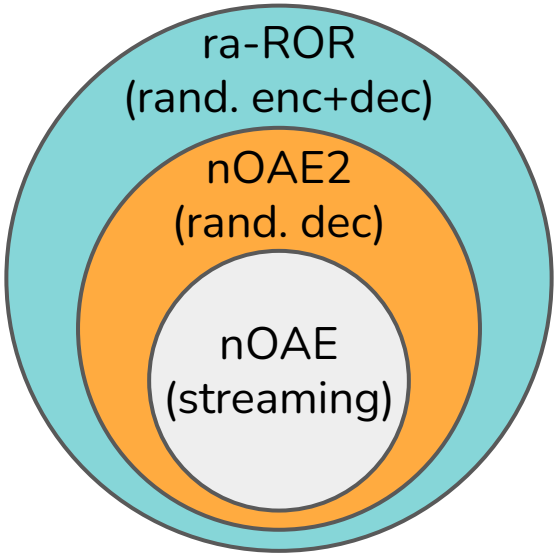

Existing schemes (e.g., STREAM, Tink streaming) remain secure under ra-ROR!
⇒ For example, safe to encrypt segment in parallel

FLOE satisfies ra-ROR security!

ra-ROR = mu-PRF security of F + mu-ROR$ security of AEAD

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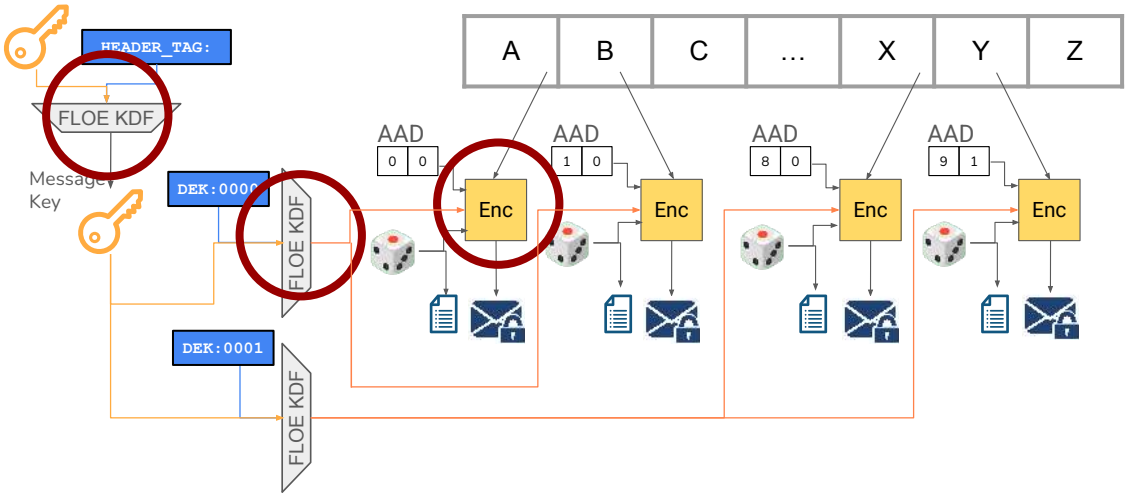

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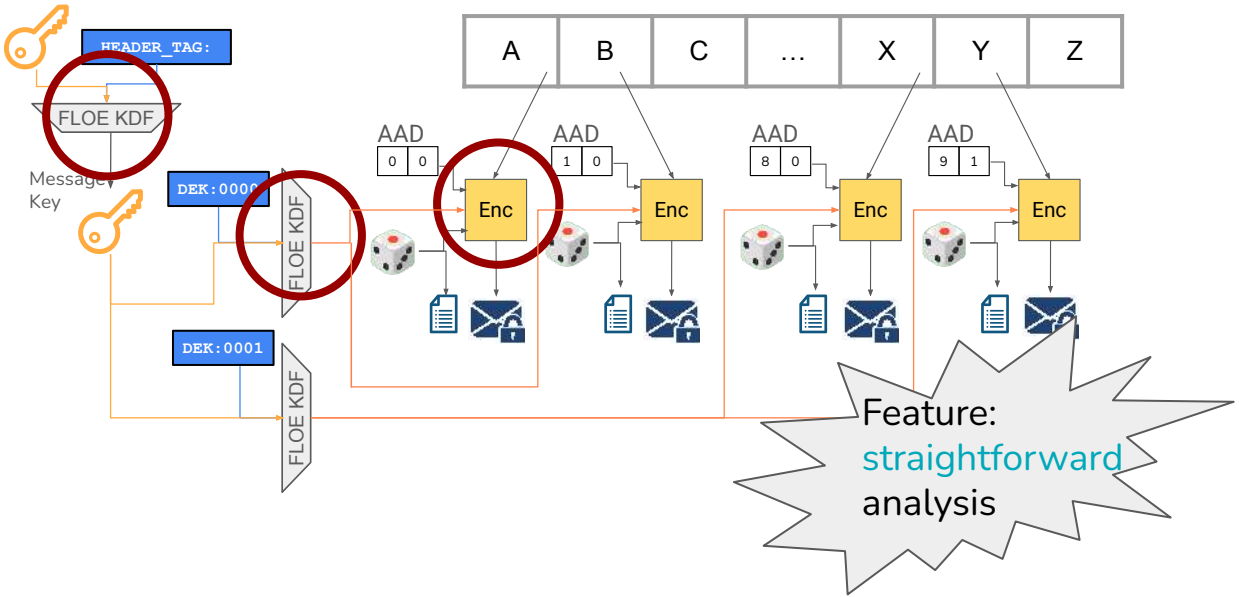

FLOE security scaling

- AES-GCM requires frequent key rotations! Random IVs: 2^{32} messages/key [KCCP24]
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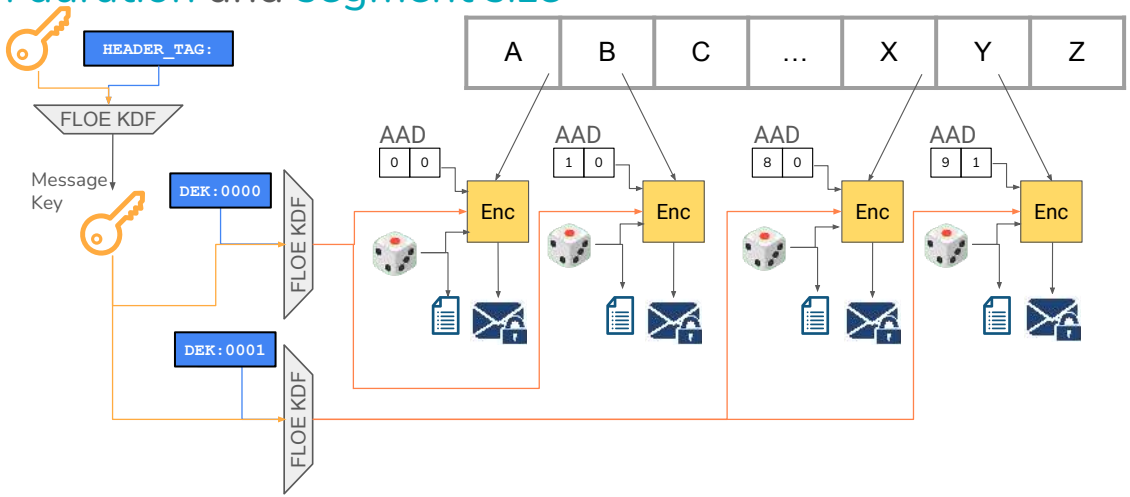

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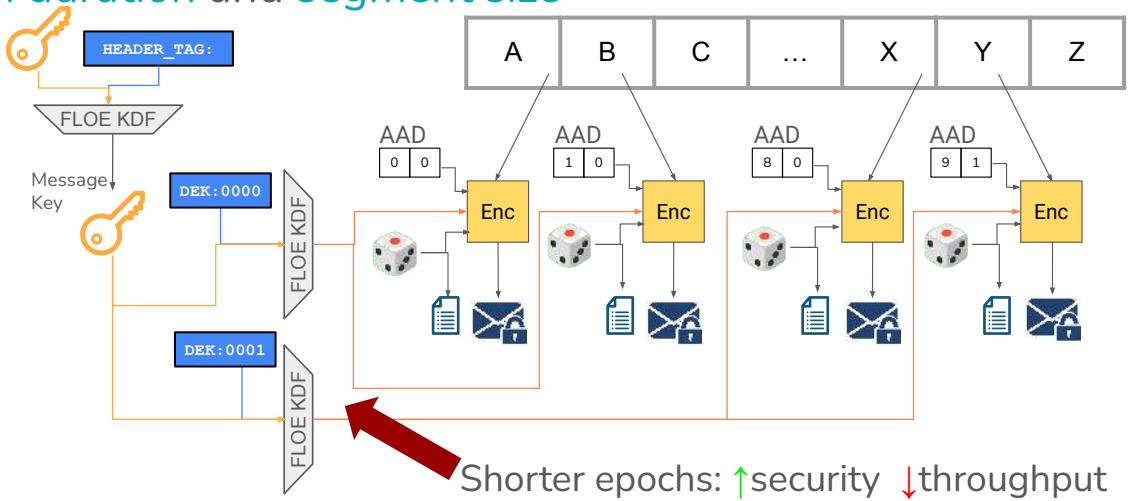

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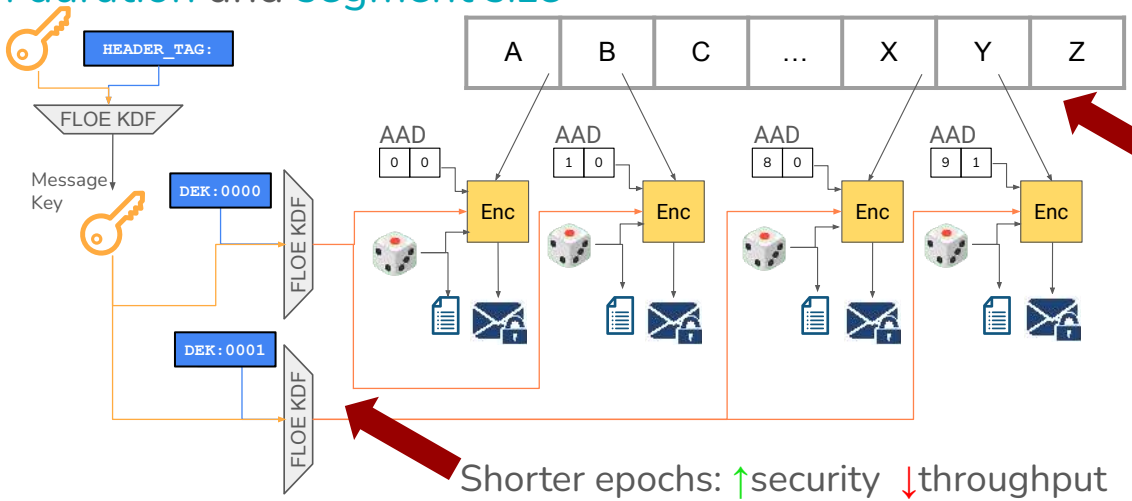

Shorter segments: ↓security ↓memory

Shorter epochs: ↑security ↓throughput

Commitment for random-access AE

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⇒ ra-CMT even using AES-GCM!

FLOE Performance

FLOE versus AES-GCM

“Streaming” Encryption

MacBook Pro, M3 Max, Java, 1GB

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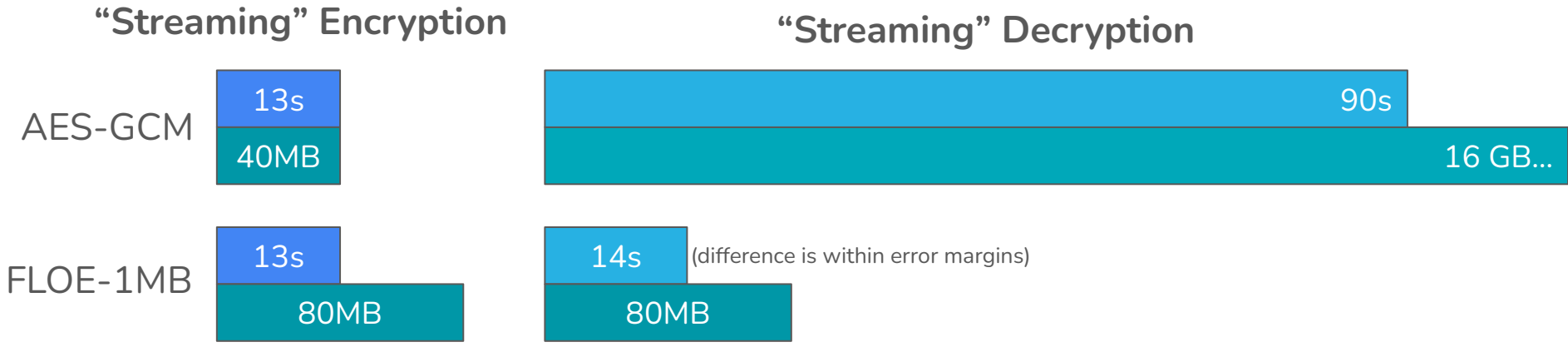

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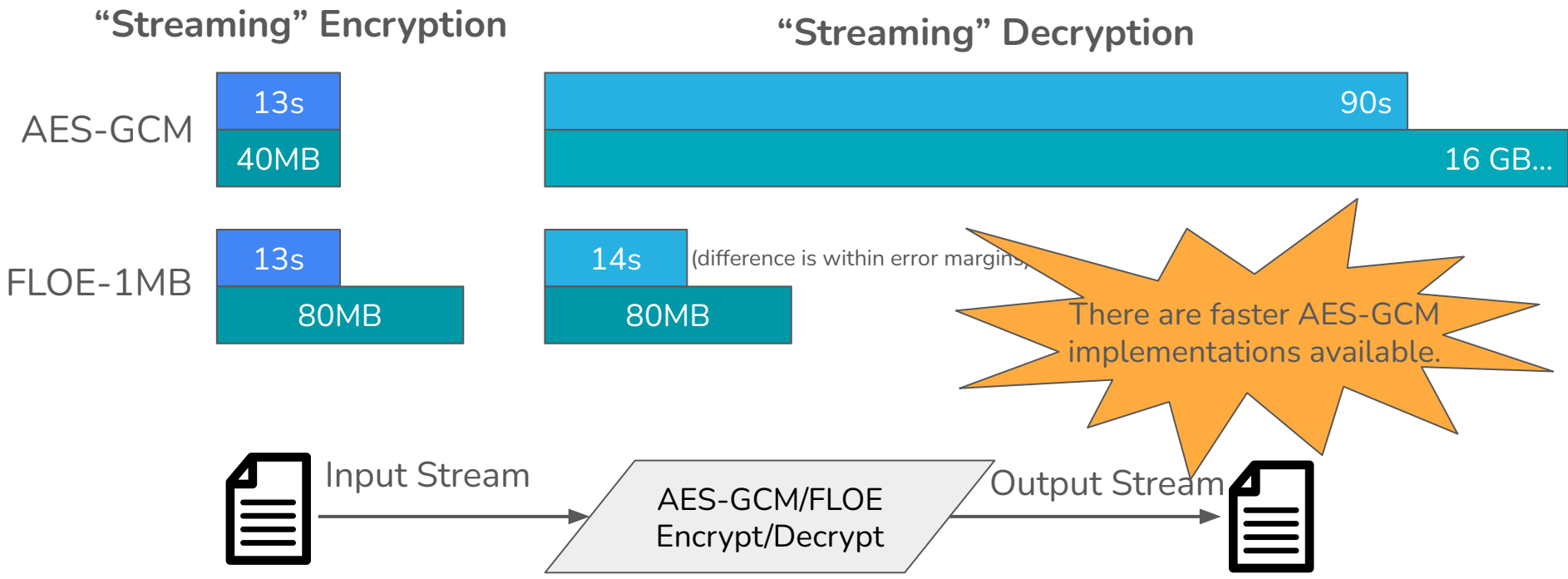

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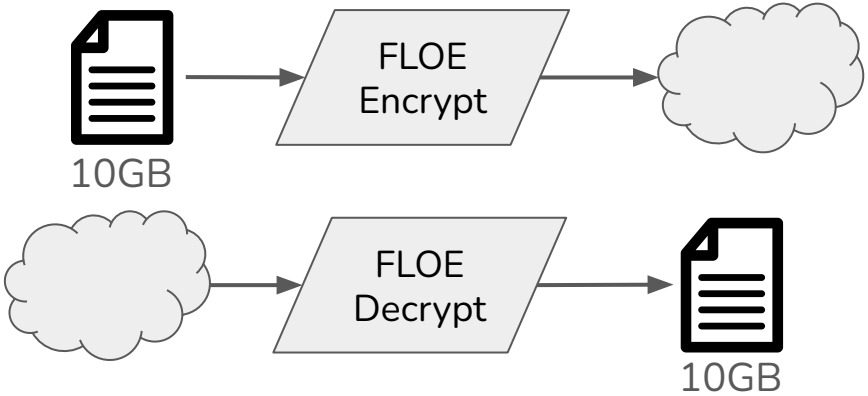
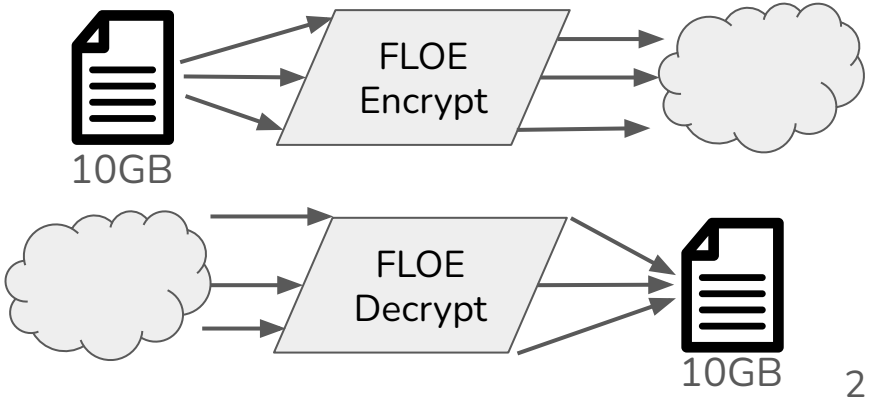
FLOE Performance

Sequential vs Multi-Threaded





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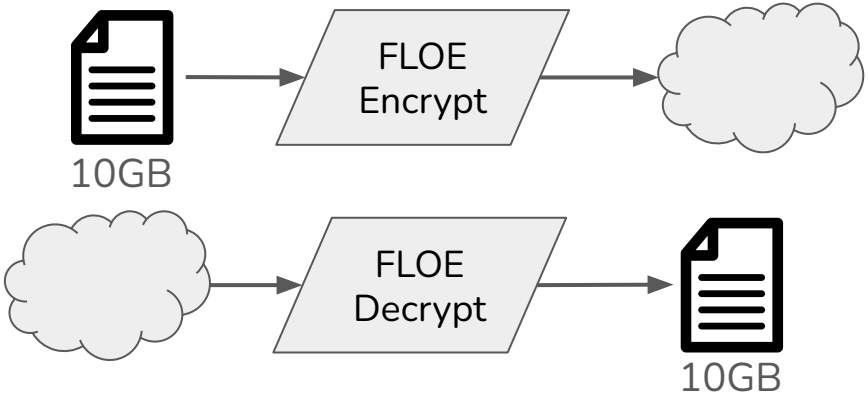

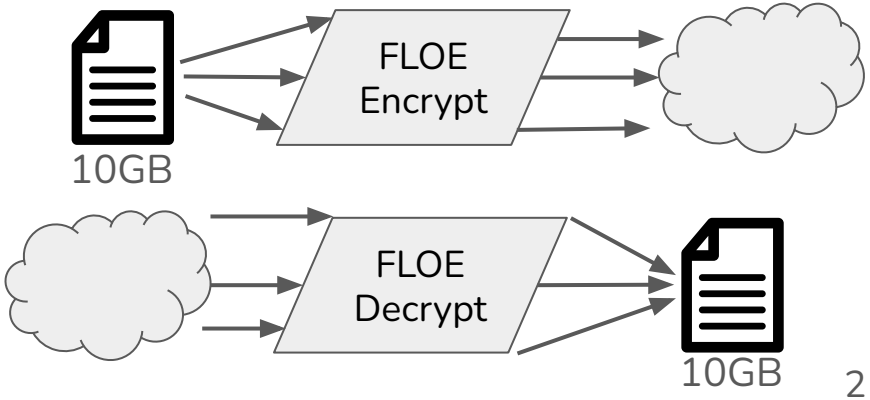

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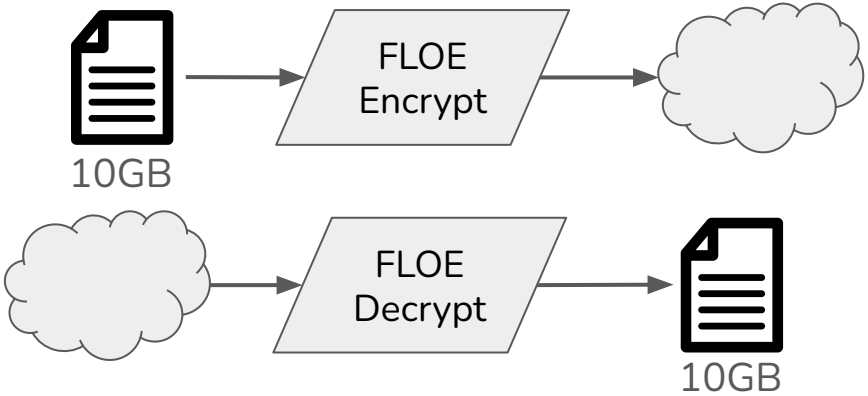
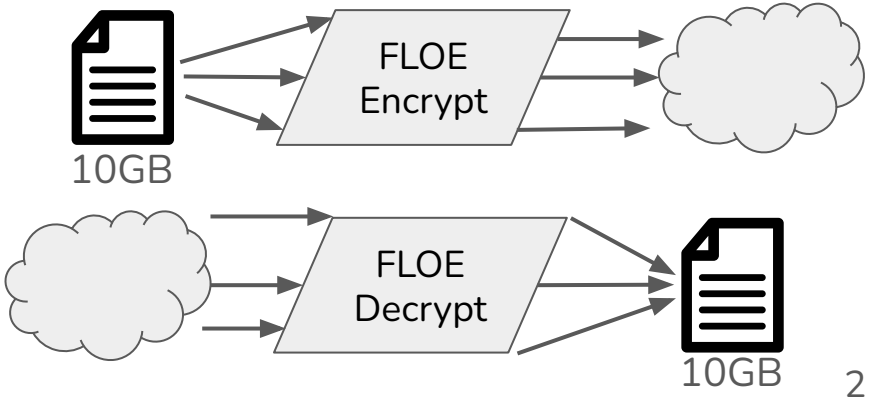
CPU-bound





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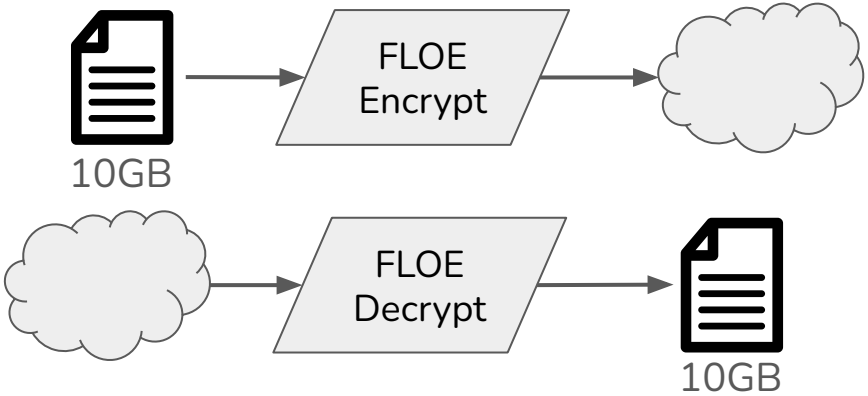
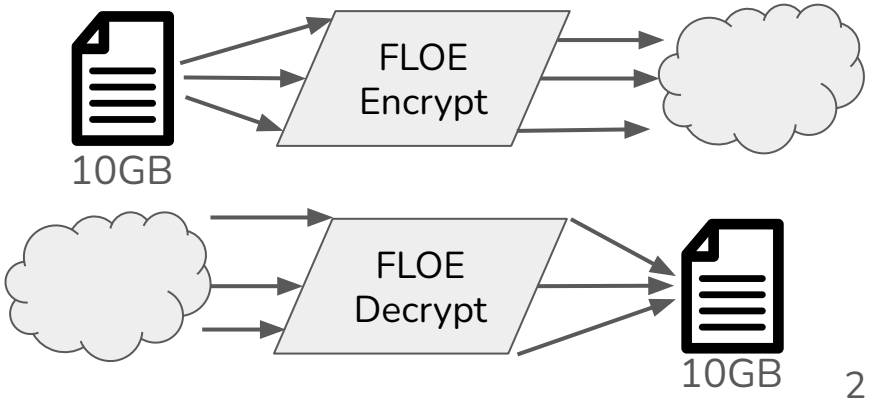


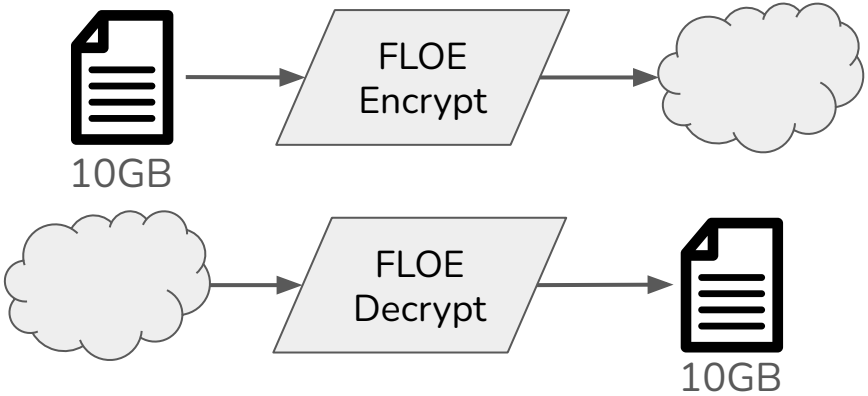


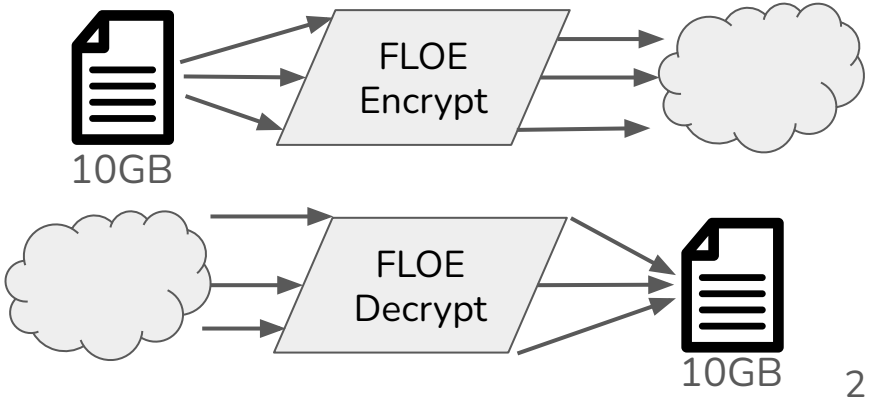
FLOE Performance

Sequential vs Multi-Threaded

Sequential — ~ 3.5 minutes — CPU-bound

Random-Access Multi-threaded — ~1.5 minutes — Network-bound, 1Gbps





FLOE Performance

Sequential vs Multi-Threaded

Faster
Lower-Resources
More Secure

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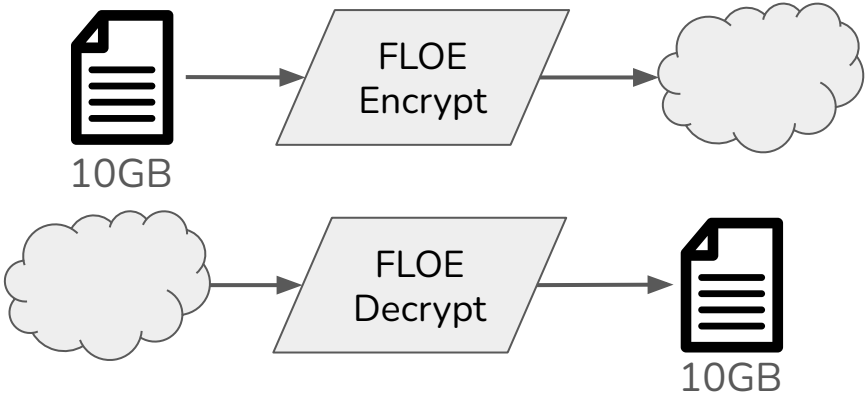

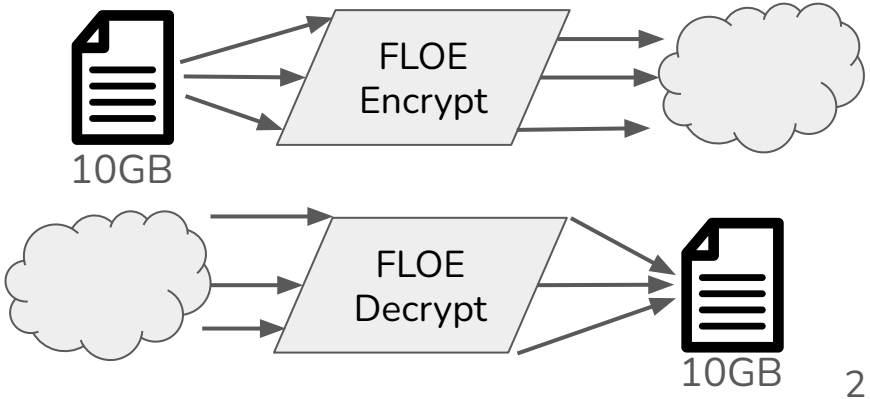

You can use FLOE now!

Public Specification

Available on GitHub

No patents

Anyone is allowed to use it

Reference Code

Available on GitHub

Apache 2.0 Licensed

Java, Go, C++, and Rust

Community Involvement

Part of the Community Cryptography Specification Project (C2SP)

Already ported to the language Zig in < 400 lines of code.

https://github.com/Snowflake-Labs/floe-specification

Thank you!

Questions?

andresfg@cs.cornell.edu
greg.rubin@snowflake.com

See our paper for more details:

- Concrete bounds
- Security under multiple error messages
- Separations between definitions
- And much more!

Paper

Reference implementations

https://eprint.iacr.org/2025/2275